



Data Analytics &
Data Engineering

KNIME Case Study #3:
Classification Model

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Waranyu Wongseree

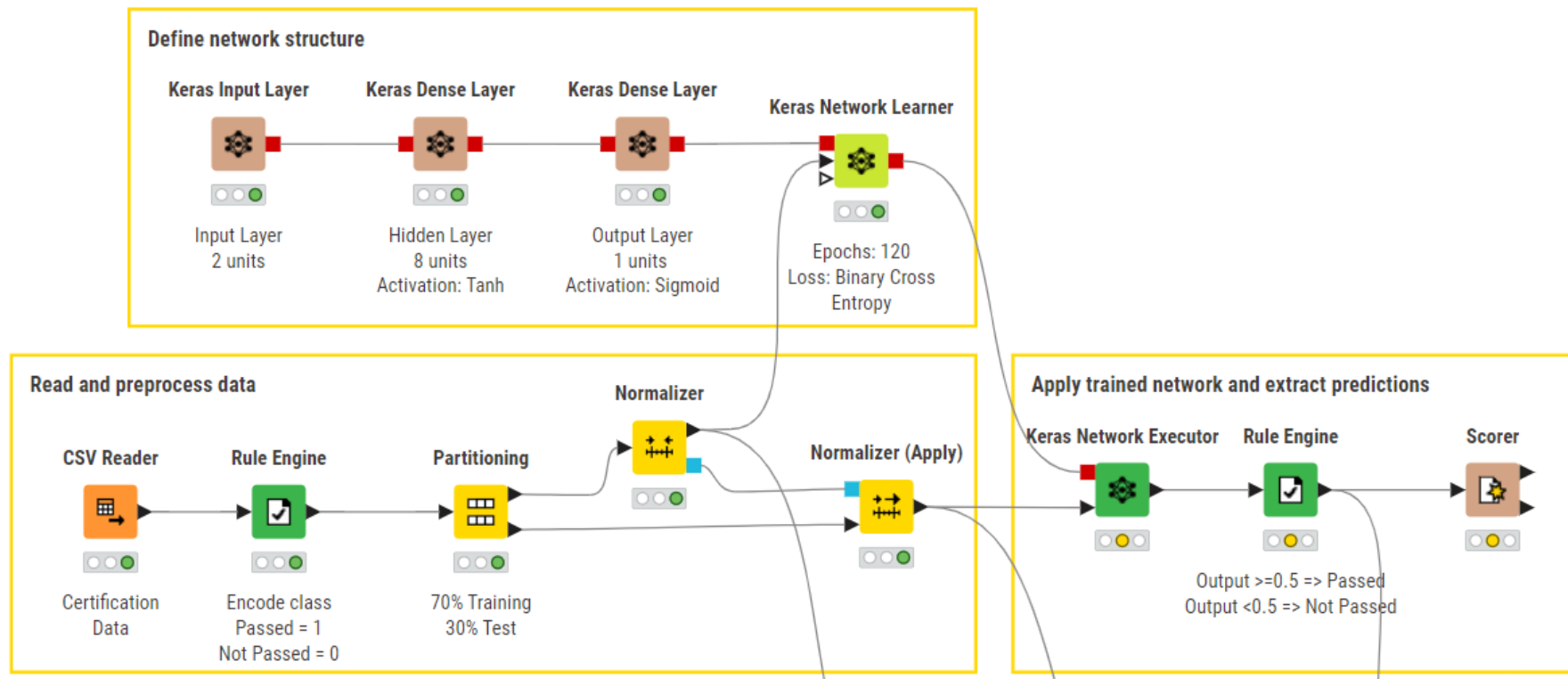
Case Study: Classification Model

- Binary Classification: Certification Data
- Multiclass classification: Iris Data
- Image classification: MNIST Data

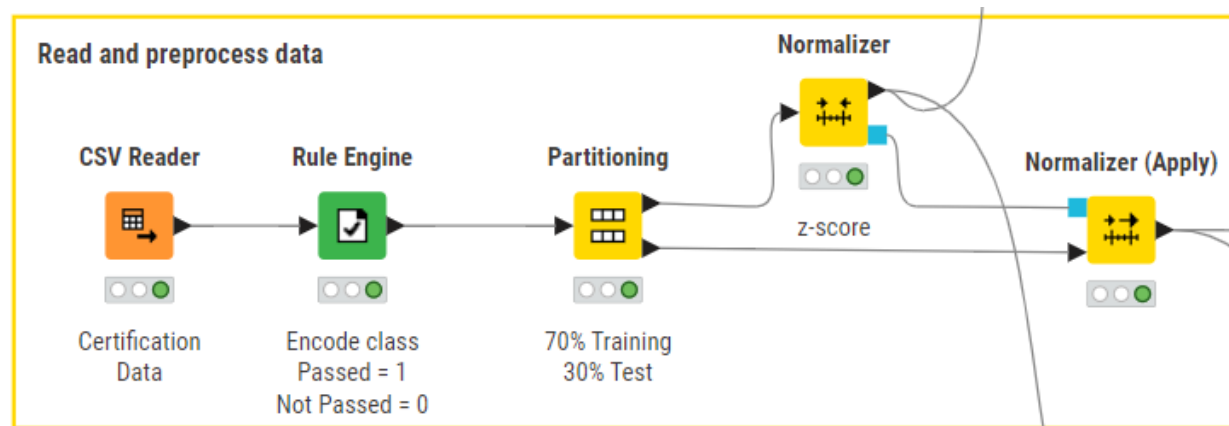
Binary Classification:

Certification Data

Multilayer Perceptron (MLP)



Part 1: Read and preprocess data



Important Setting: Rule Engine

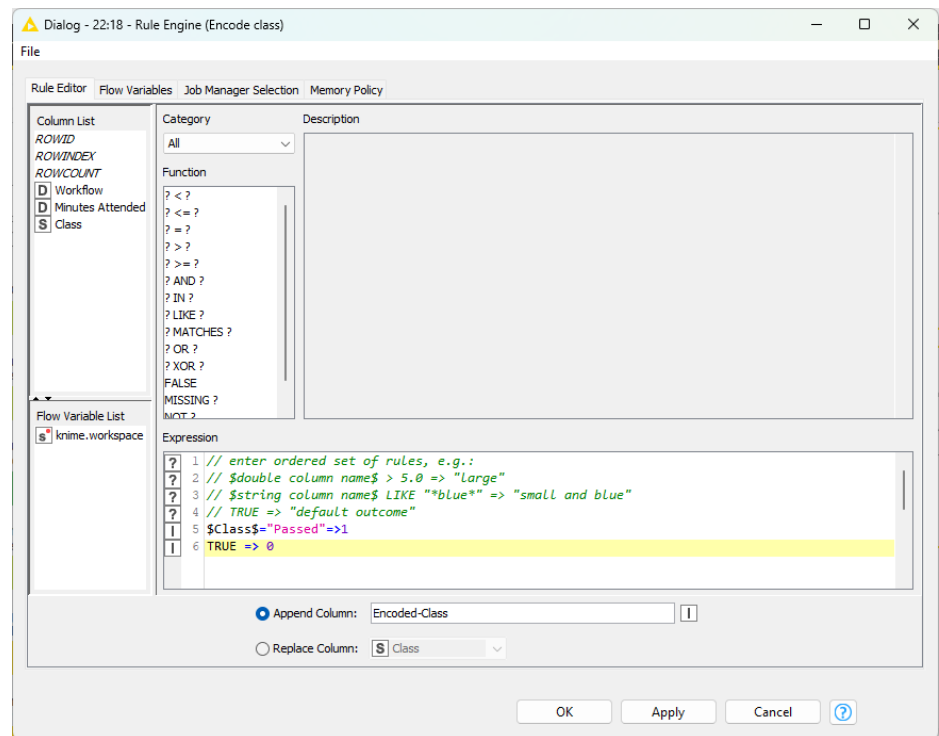
Rule Engine



Encode class
Passed = 1
Not Passed = 0

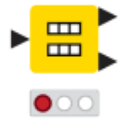
`$Class$="Passed"=>1`

`TRUE => 0`



Important Setting: Partitioning

Partitioning



70% Training
30% Test

Dialog - 23:3 - Partitioning

File

First partition | Flow Variables | Job Manager Selection | Memory Policy

Choose size of first partition

☐ Absolute 100

☒ Relative[%] 70

☐ Take from top

☐ Linear sampling

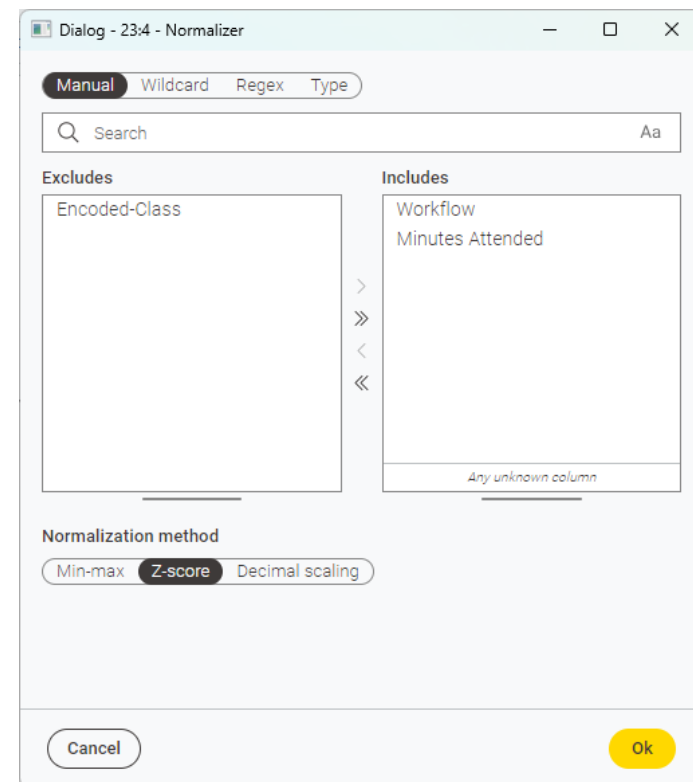
☒ Draw randomly

☐ Stratified sampling S Class

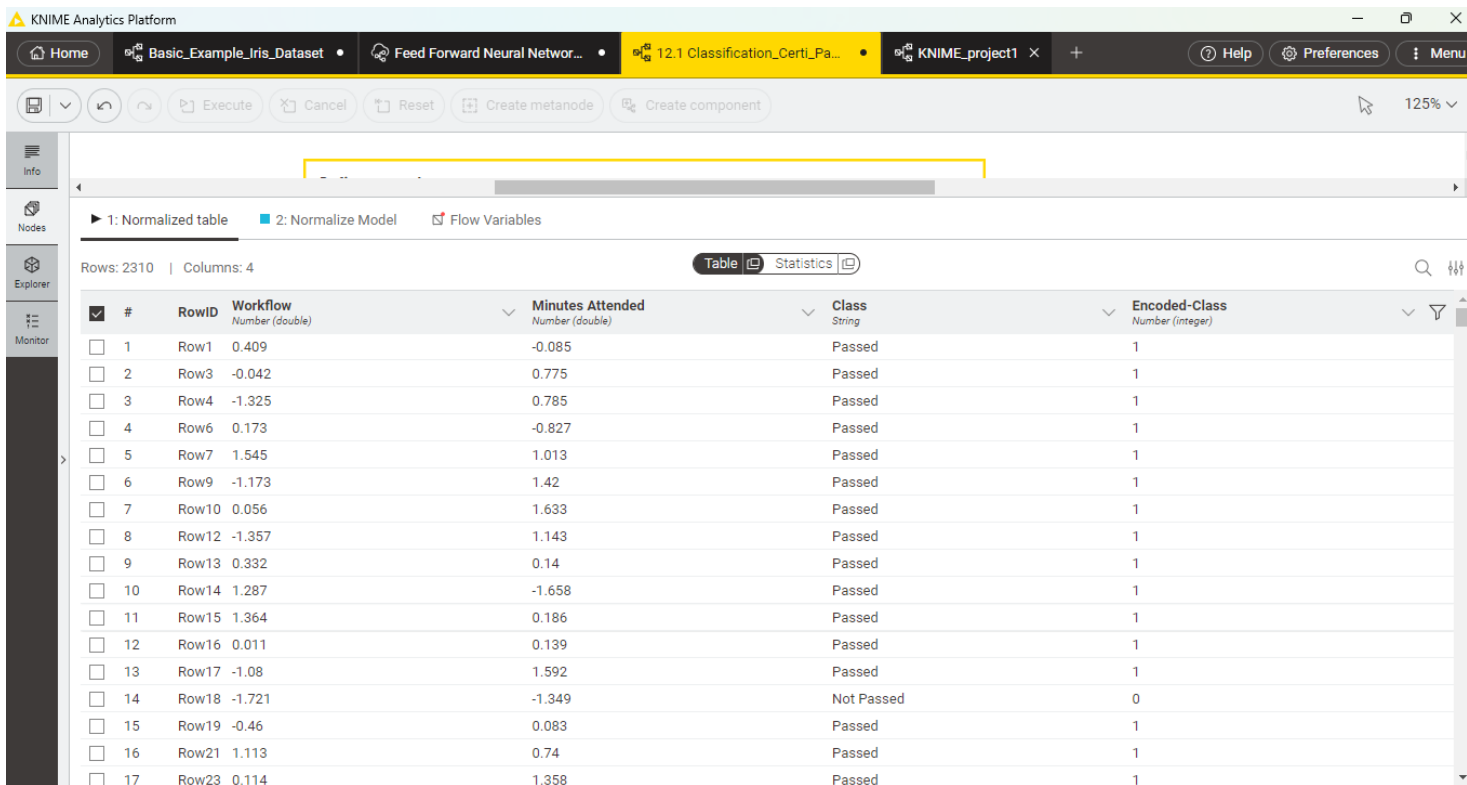
☒ Use random seed 0

OK Apply Cancel ?

Important Setting: Normalizer



Part 1: Read and preprocess data



KNIME Analytics Platform

Home Basic_Example_Iris_Dataset Feed Forward Neural Network... 12.1 Classification_Certification... KNIME_project1 x Help Preferences Menu

Execute Cancel Reset Create metanode Create component 125%

Info

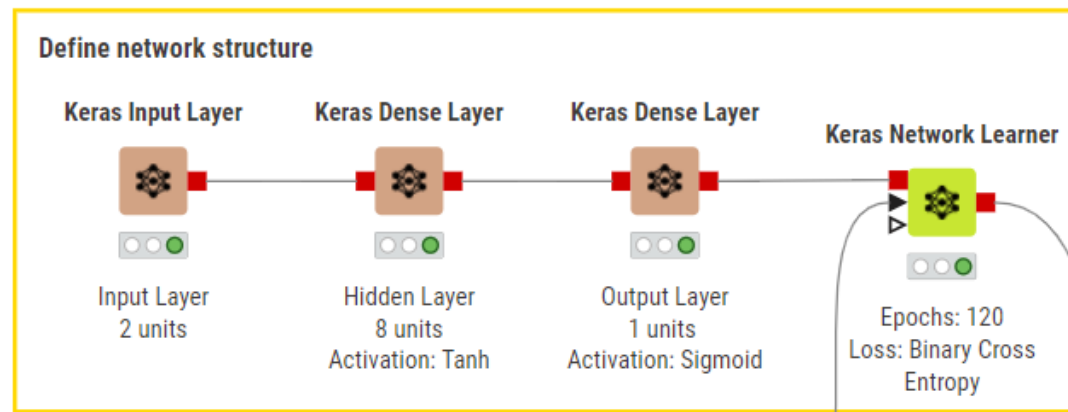
Nodes

1: Normalized table 2: Normalize Model Flow Variables

Rows: 2310 | Columns: 4 Table Statistics

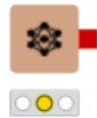
☒	#	RowID	Workflow Number (double)	Minutes Attended Number (double)	Class String	Encoded-Class Number (integer)
☐	1	Row1	0.409	-0.085	Passed	1
☐	2	Row3	-0.042	0.775	Passed	1
☐	3	Row4	-1.325	0.785	Passed	1
☐	4	Row6	0.173	-0.827	Passed	1
☐	5	Row7	1.545	1.013	Passed	1
☐	6	Row9	-1.173	1.42	Passed	1
☐	7	Row10	0.056	1.633	Passed	1
☐	8	Row12	-1.357	1.143	Passed	1
☐	9	Row13	0.332	0.14	Passed	1
☐	10	Row14	1.287	-1.658	Passed	1
☐	11	Row15	1.364	0.186	Passed	1
☐	12	Row16	0.011	0.139	Passed	1
☐	13	Row17	-1.08	1.592	Passed	1
☐	14	Row18	-1.721	-1.349	Not Passed	0
☐	15	Row19	-0.46	0.083	Passed	1
☐	16	Row21	1.113	0.74	Passed	1
☐	17	Row23	0.114	1.358	Passed	1

Part 2: Define network structure



Important Setting: Keras Input Layer

Keras Input Layer



Input Layer
2 units

Dialog - 22:15 - Keras Input Layer (Input Layer)

File

Options | Flow Variables | Job Manager Selection

Name prefix ☐

Shape

Batch size ☐

Data type

Data format

OK Apply Cancel ?

Important Setting: Keras Dense Layer

Keras Dense Layer



Hidden Layer
8 units
Activation: Tanh

Dialog - 23:7 - Keras Dense Layer

File

Options Advanced Flow Variables Job Manager Selection

Name prefix ☐

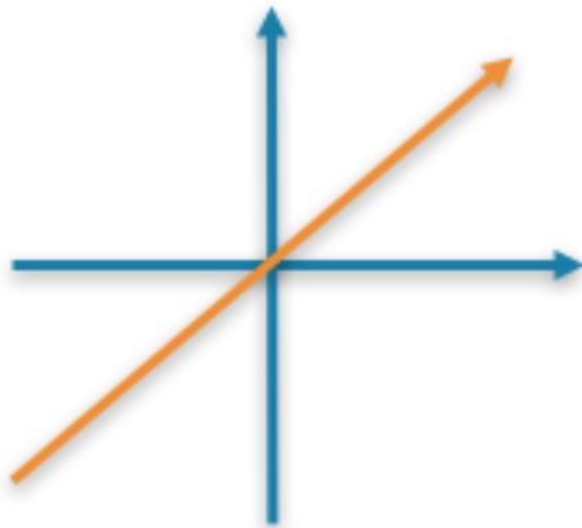
Input tensor input_1_0:0 [2] float

Units 8

Activation function Tanh

OK Apply Cancel ?

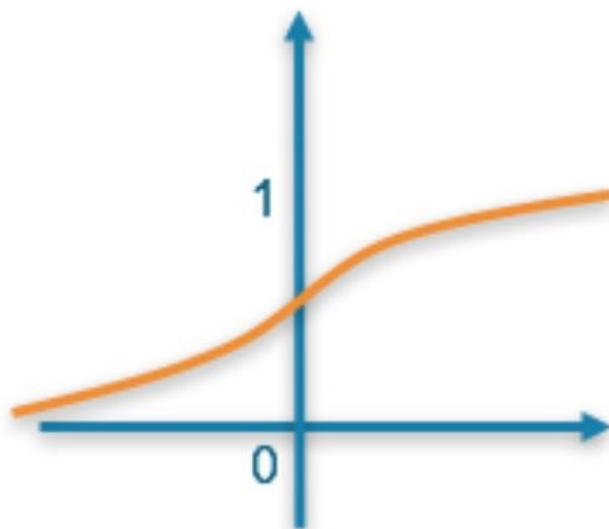
Linear Activation Function



$$f(z) = az$$

Range: -infinity to infinity

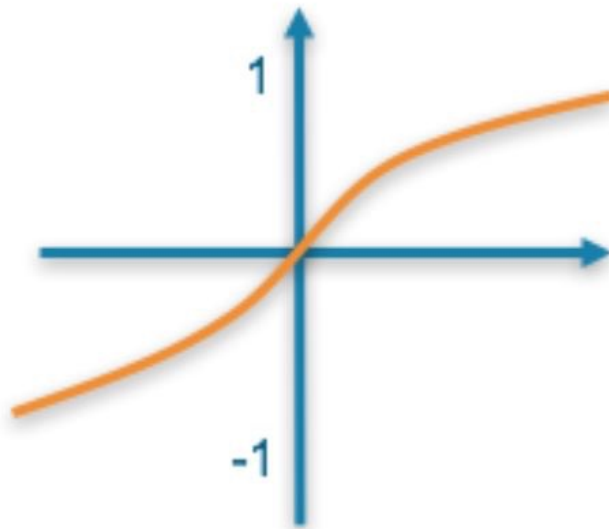
Sigmoid Activation Function



$$\text{sigmoid}(z) = \frac{1}{1 + e^{-z}}$$

Range: 0 to 1

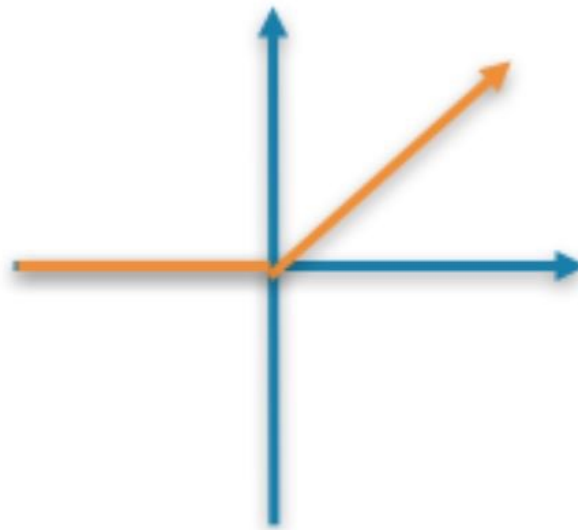
Hyperbolic Tangent Activation |



$$\tanh(z) = \frac{e^z - e^{-z}}{e^z + e^{-z}}$$

Range: -1 to 1

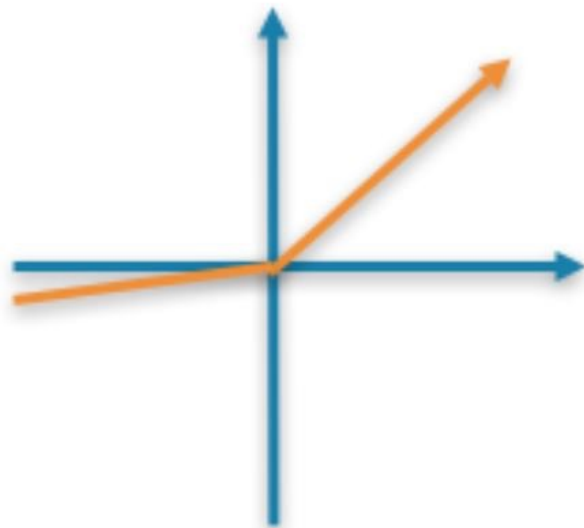
Rectified Linear Unit (ReLU)



$$\text{relu}(z) = \max(0, z)$$

Range: 0 to infinity

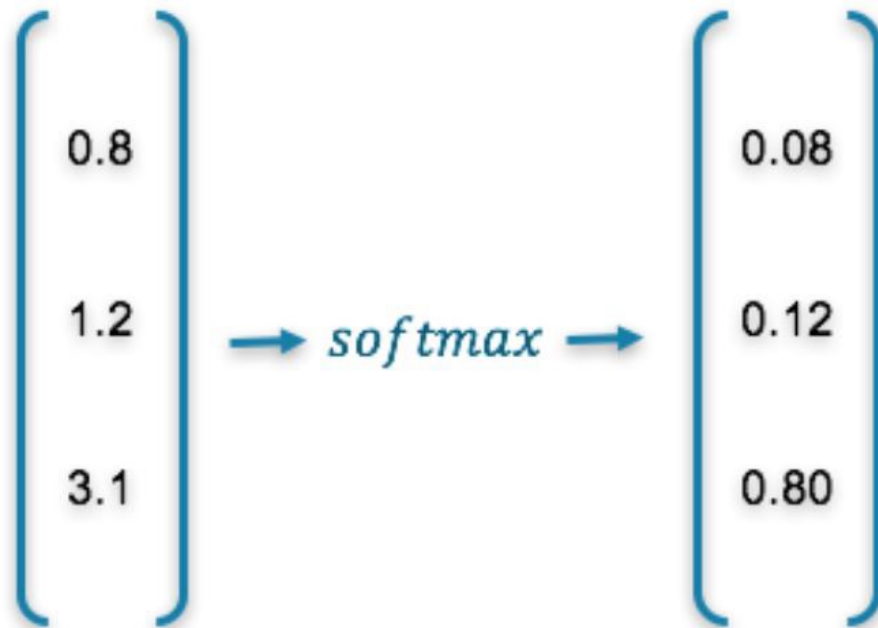
Leaky ReLU Activation Function



$$\text{leakyrelu}(z) = \begin{cases} 0.01z & \text{for } z < 0 \\ z & \text{for } z \geq 0 \end{cases}$$

Range: -infinity to infinity

Softmax Activation Function



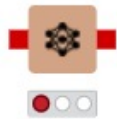
$$\text{softmax}(z_i) = \frac{\exp(z_i)}{\sum_j \exp(z_j)}$$

Range: 0 to 1

Divides output so that the total sum of the output is equal to 1

Important Setting: Keras Dense Layer

Keras Dense Layer



Output Layer
1 units
Activation: Sigmoid

Dialog - 22:17 - Keras Dense Layer (Output Layer)

File

Options Advanced Flow Variables Job Manager Selection

Name prefix ☒ Output

Input tensor dense_1_0:0 [8] float

Units 1

Activation function Sigmoid

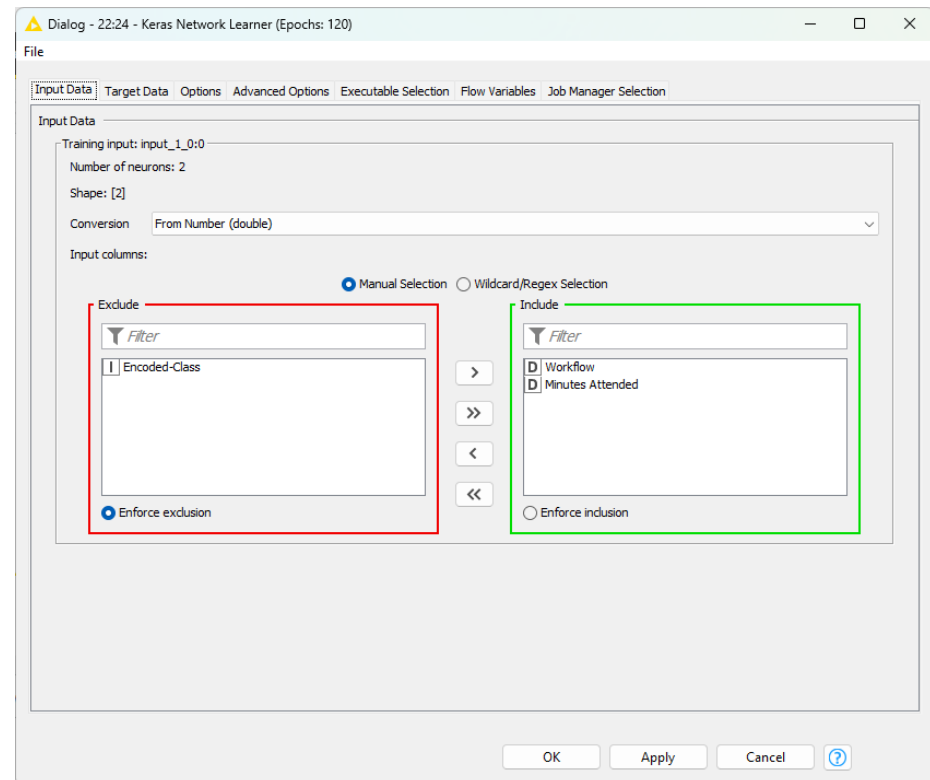
OK Apply Cancel ?

Important Setting: Keras Network Learner

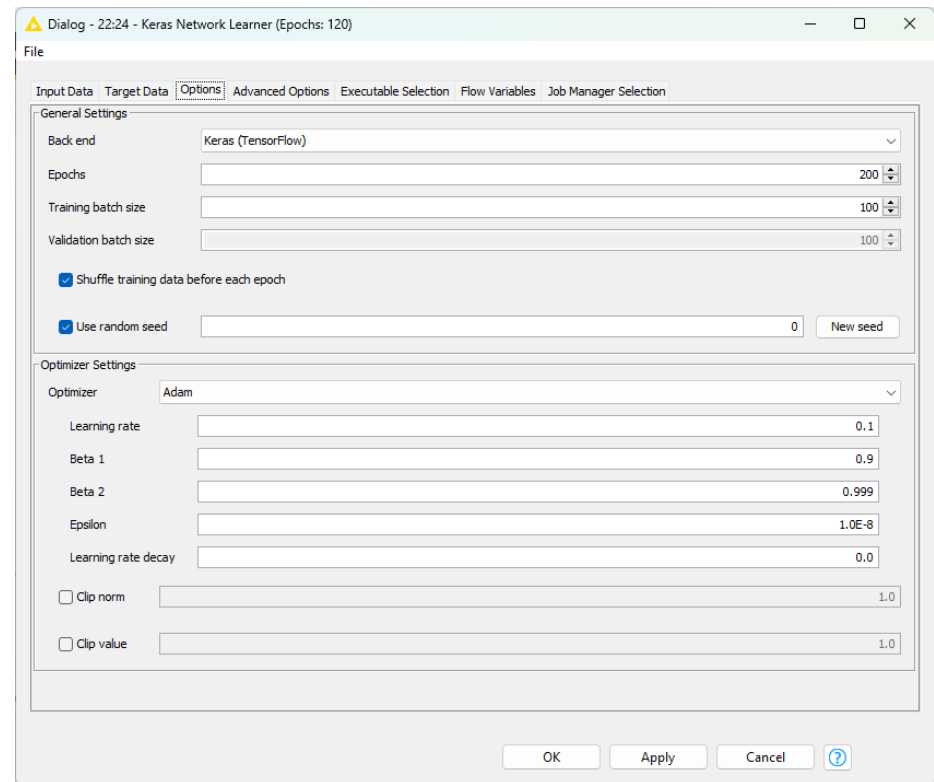
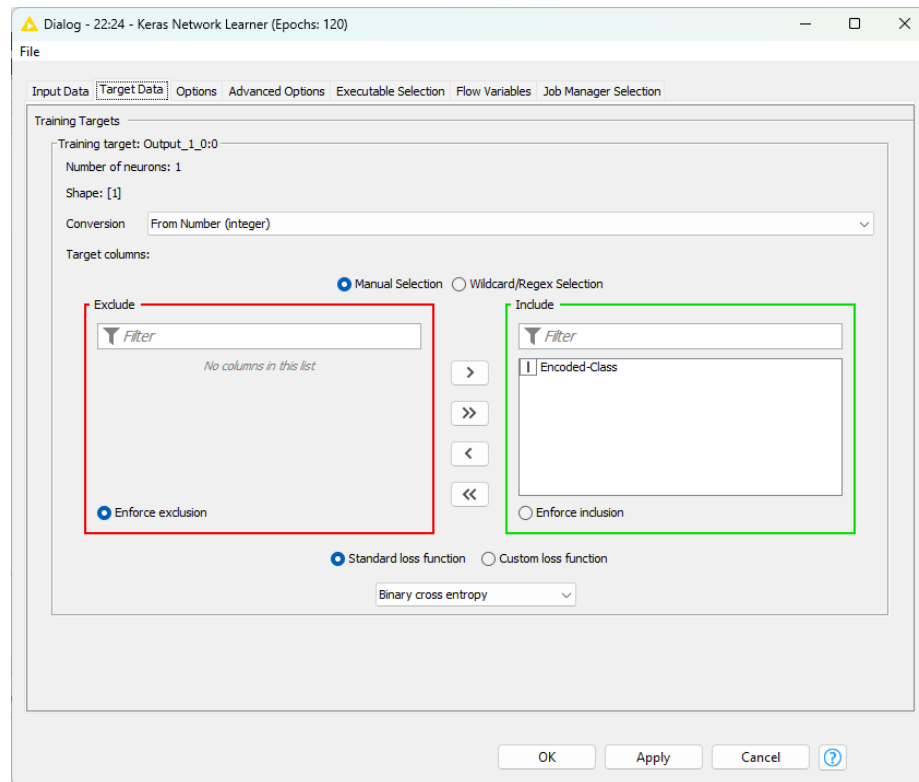
Keras Network Learner



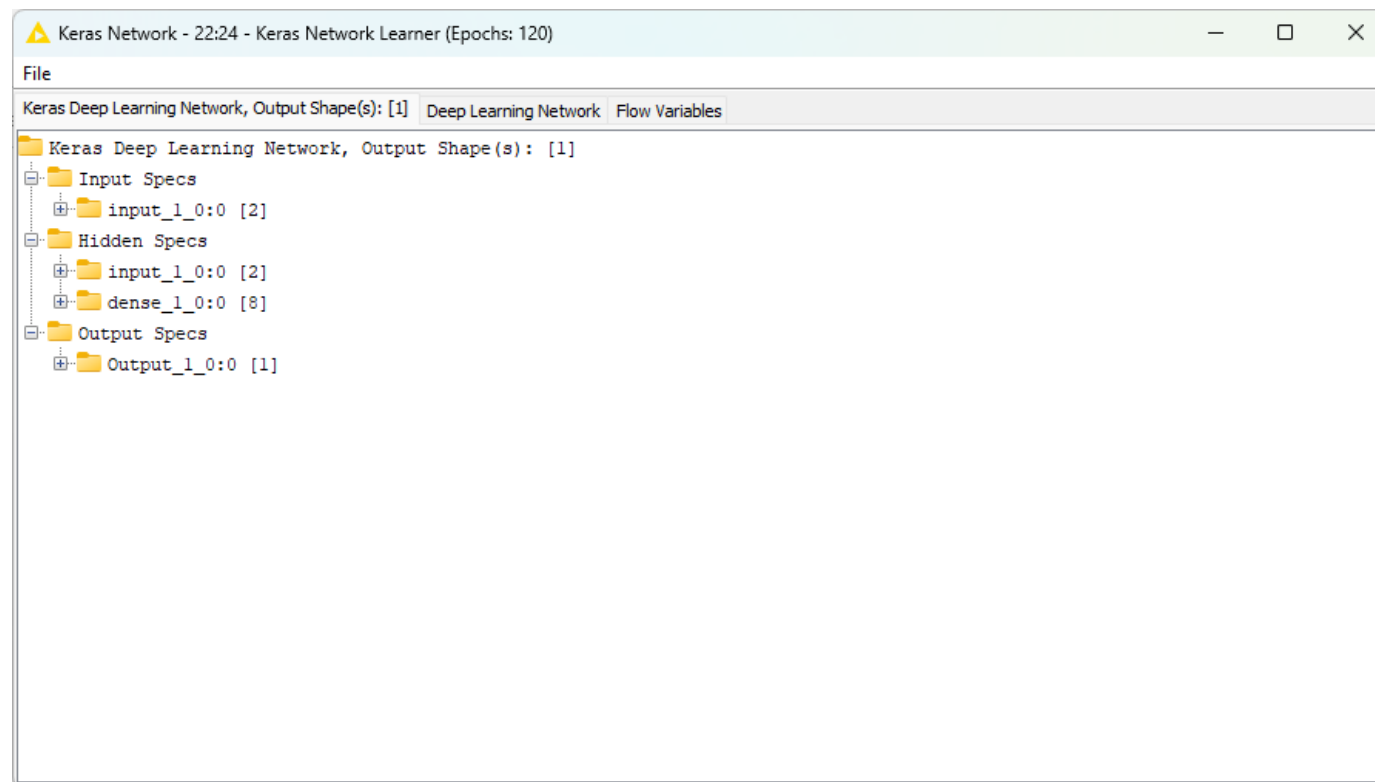
Epochs: 120
Loss: Binary Cross
Entropy



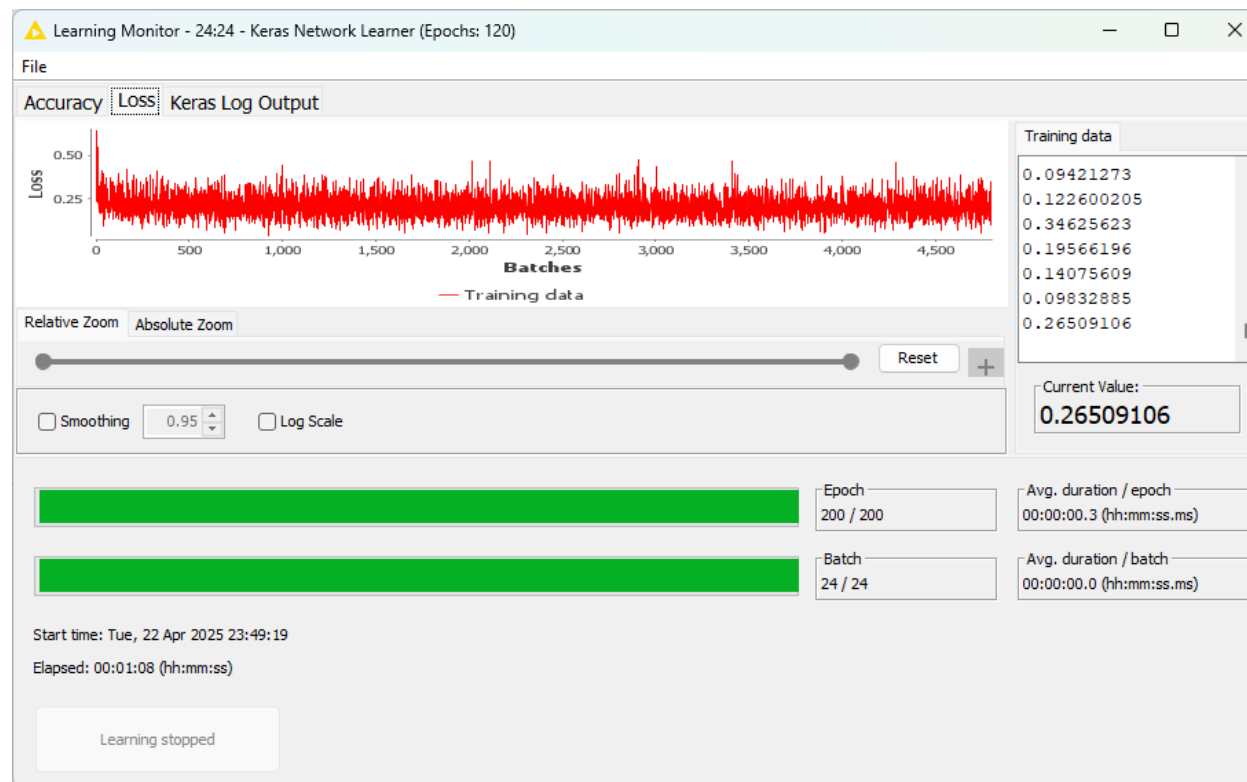
Important Setting: Keras Network Learner



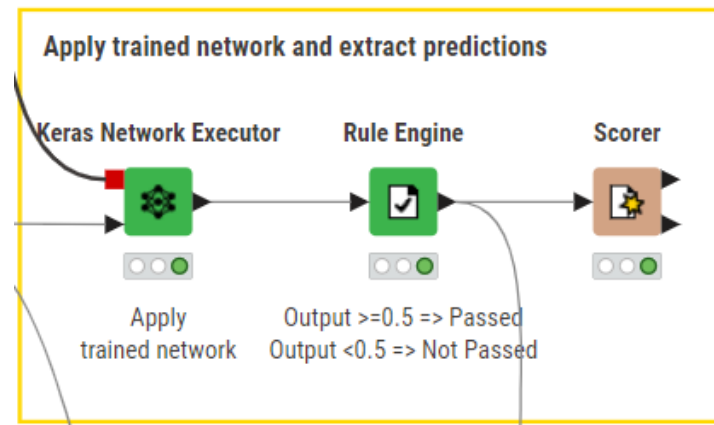
Part 2: Define network structure



Part 2: Define network structure



Part 3: Apply trained network and extract predictions

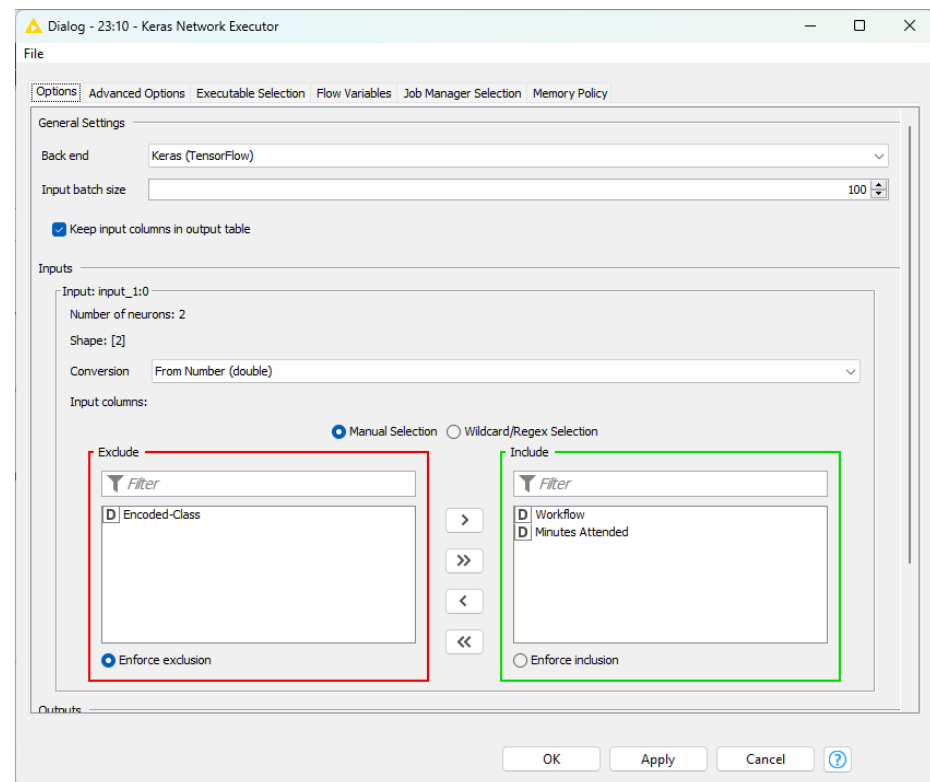


Important Setting: Keras Network Executor

Keras Network Executor



Apply
trained network

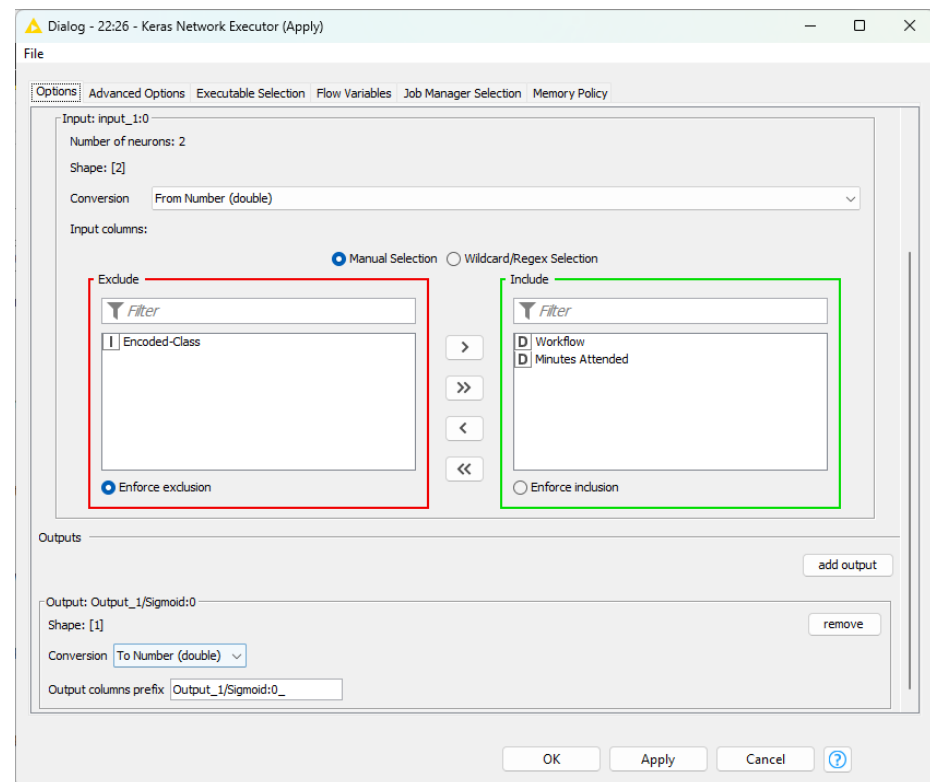


Important Setting: Keras Network Executor

Keras Network Executor



Apply
trained network



Important Setting: Rule Engine

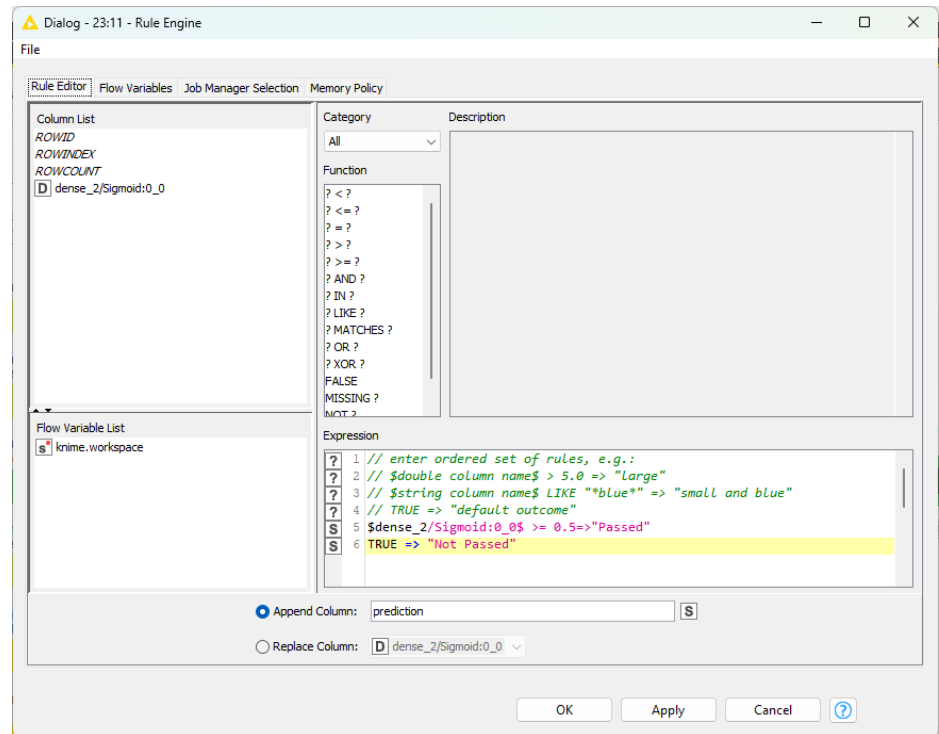
Rule Engine



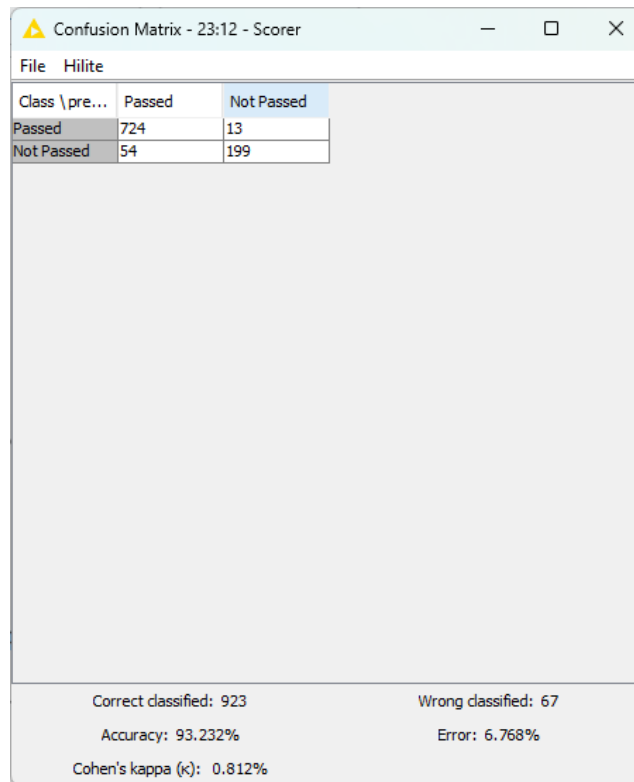
Output $\geq 0.5 \Rightarrow$ Passed
Output $< 0.5 \Rightarrow$ Not Passed

$\$dense_2/Sigmoid:0_0\$ \geq 0.5 \Rightarrow$ "Passed"

TRUE $\Rightarrow$ "Not Passed"



Part 3: Apply trained network and extract predictions



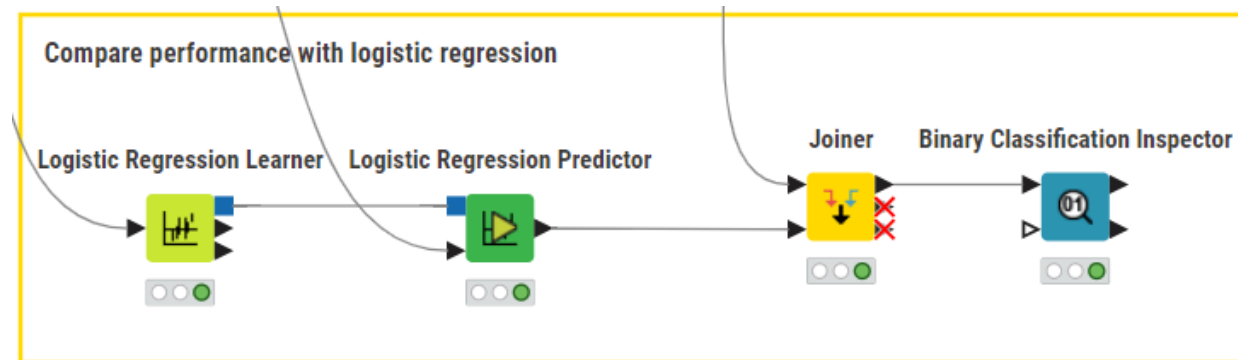
A screenshot of a software window titled "Confusion Matrix - 23:12 - Scorer". The window contains a confusion matrix table and summary statistics at the bottom.

Class \ pre...	Passed	Not Passed
Passed	724	13
Not Passed	54	199

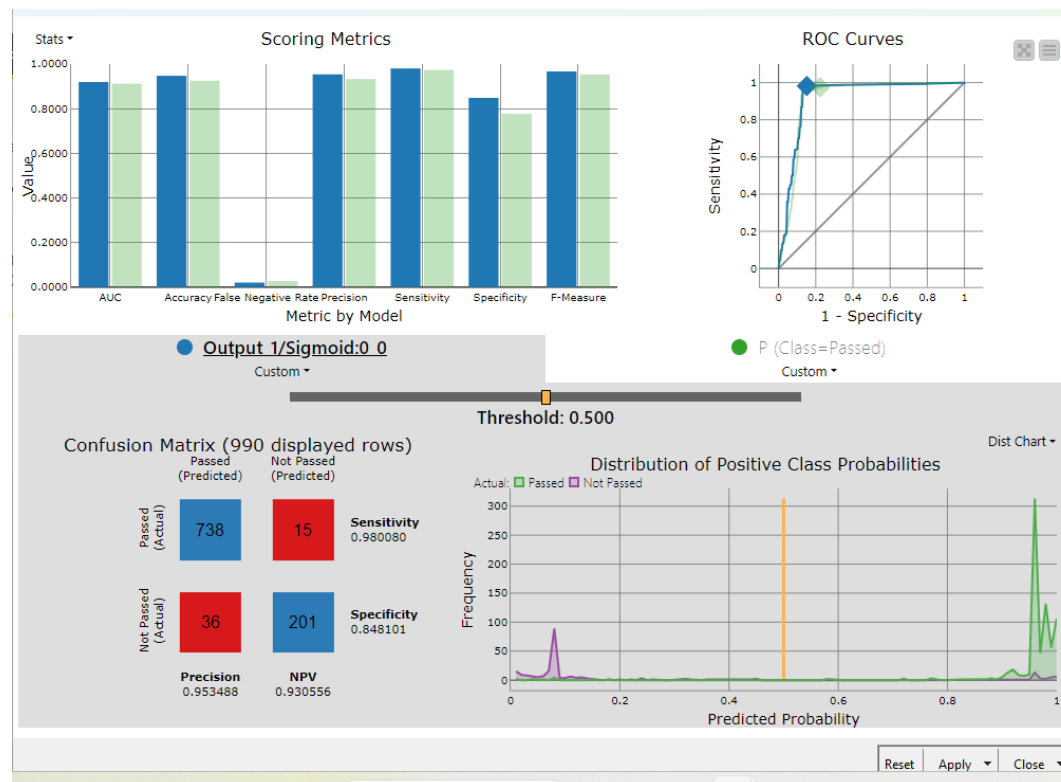
Summary statistics:

- Correct classified: 923
- Wrong classified: 67
- Accuracy: 93.232%
- Error: 6.768%
- Cohen's kappa (κ): 0.812%

Part 4: Compare performance with logistic regression



Part 4: Compare performance with logistic regression

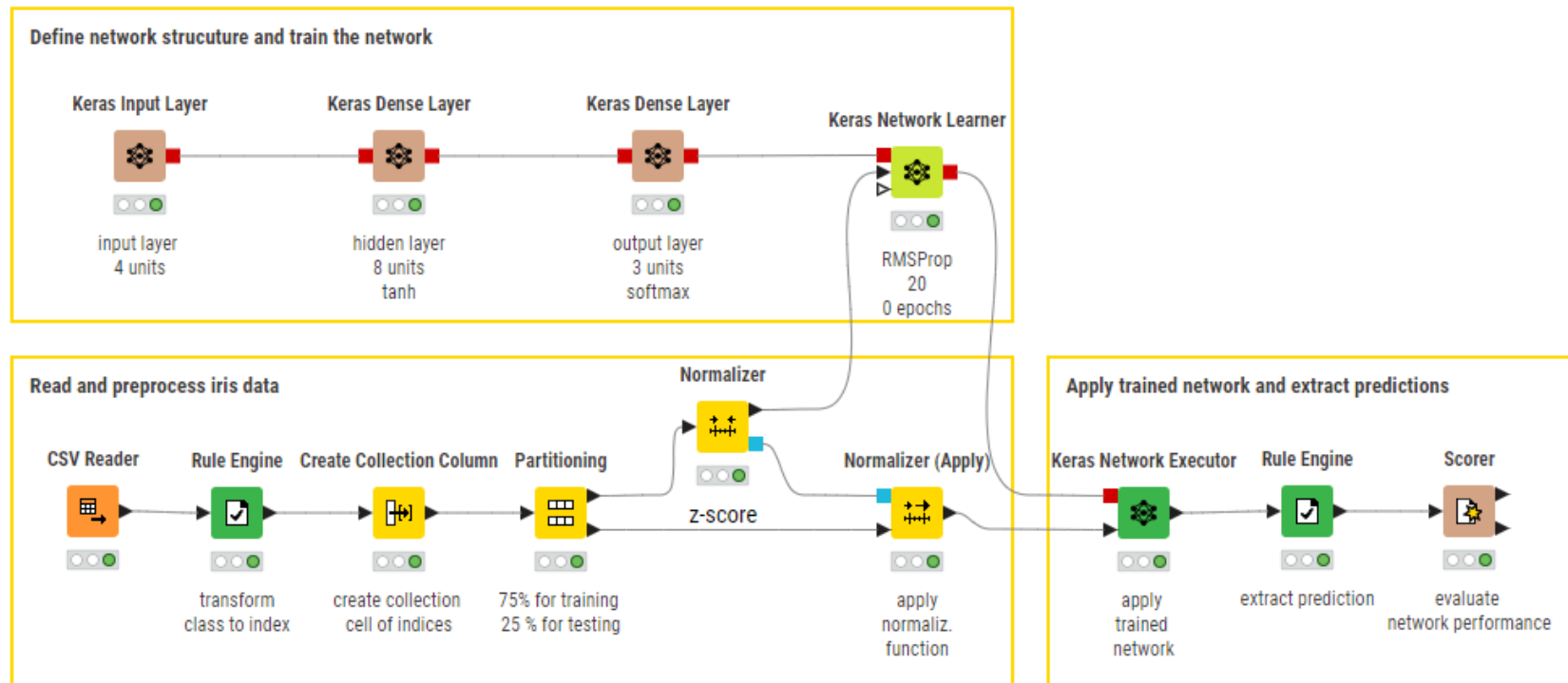


Multiclass classification:

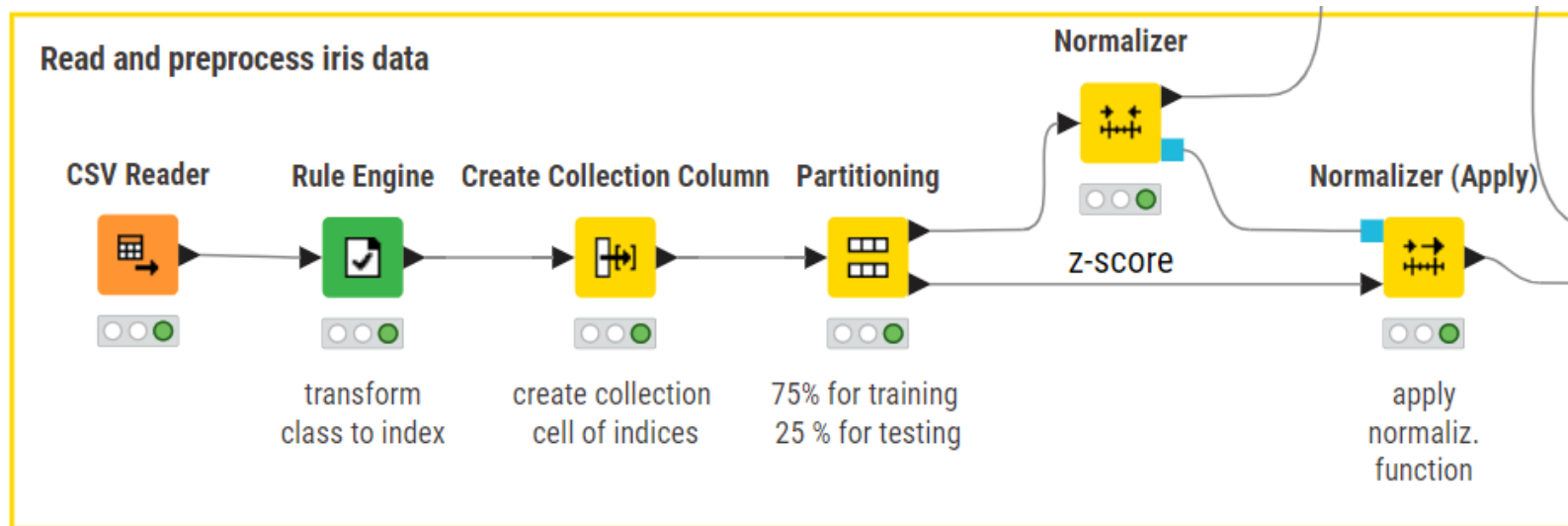
Iris Data



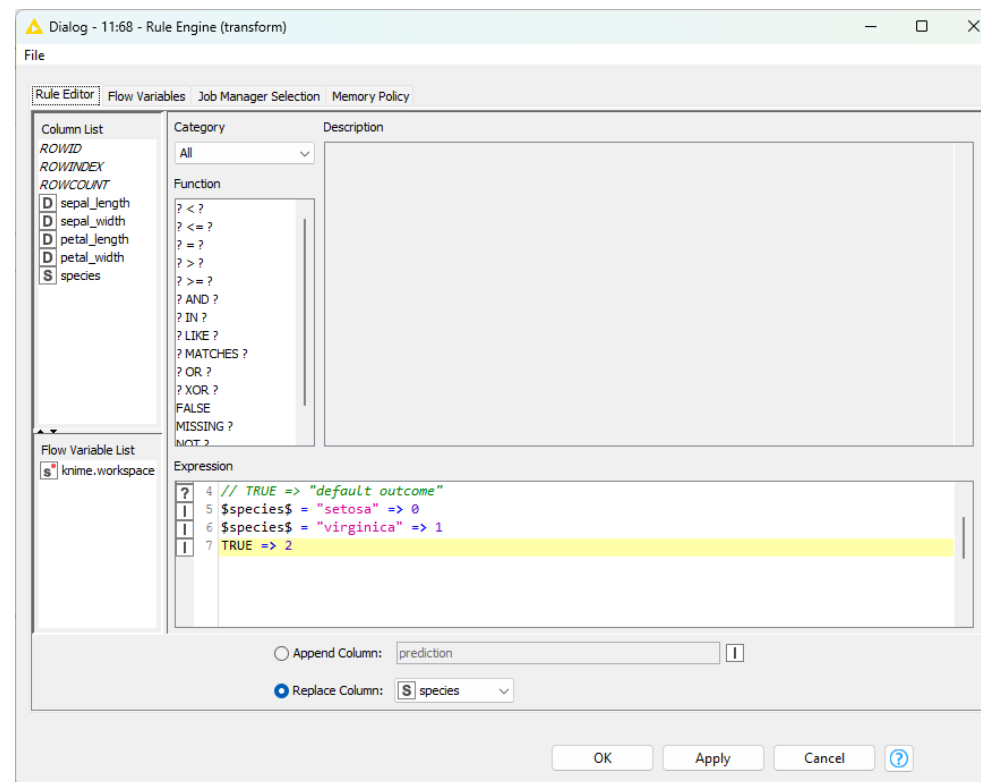
Multilayer Perceptron (MLP)



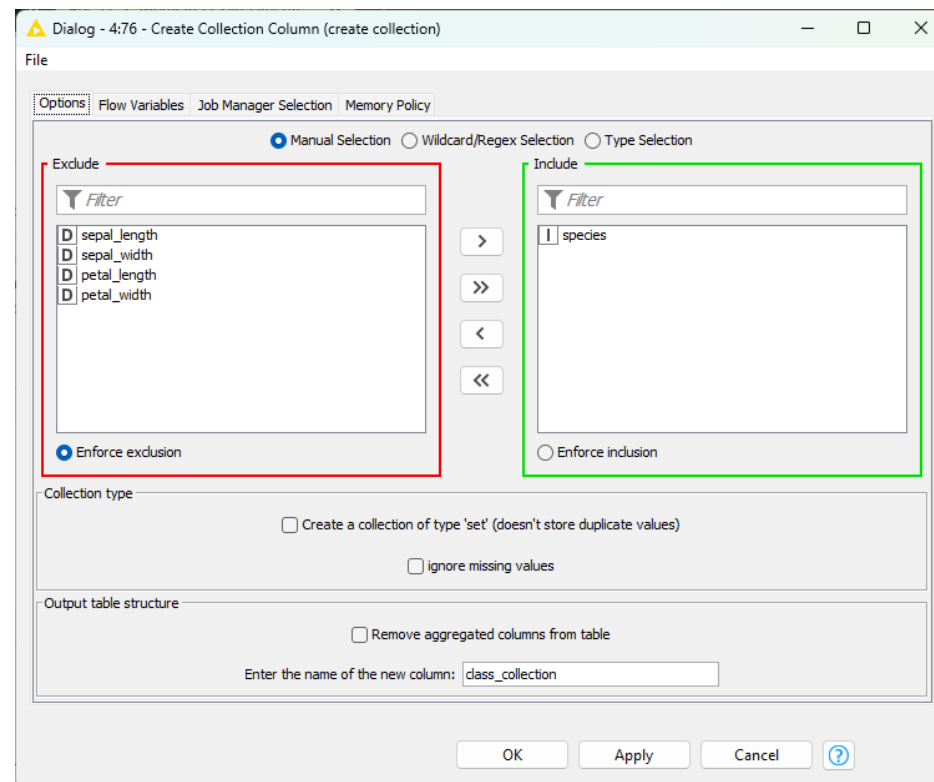
Part 1: Read and preprocess data



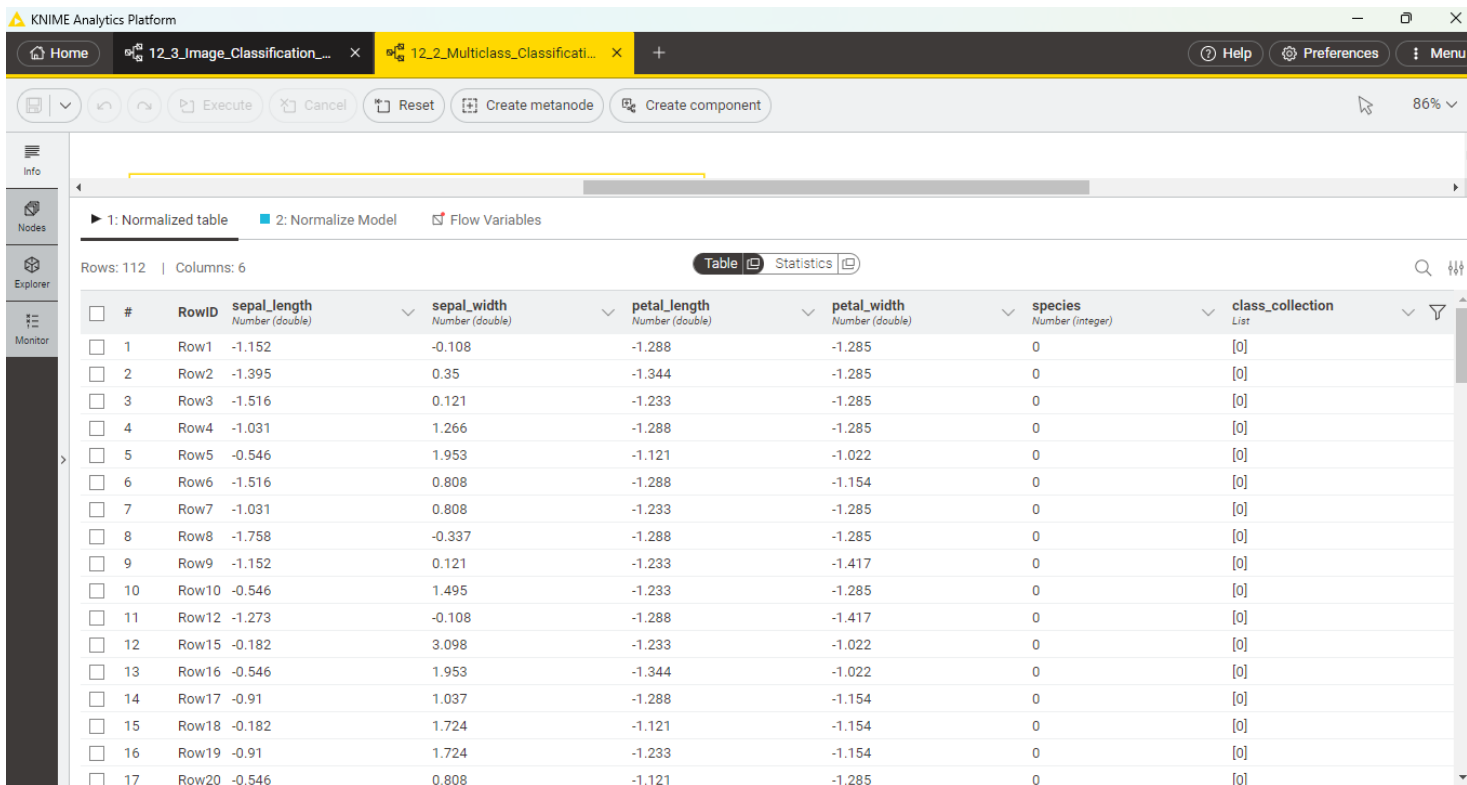
Important Setting: Rule Engine



Important Setting: Create Collection Column



Part 1: Read and preprocess data



KNIME Analytics Platform

Home 12_3_Image_Classification_... 12_2_Multiclass_Classificati... + Help Preferences Menu

Execute Cancel Reset Create metanode Create component 86%

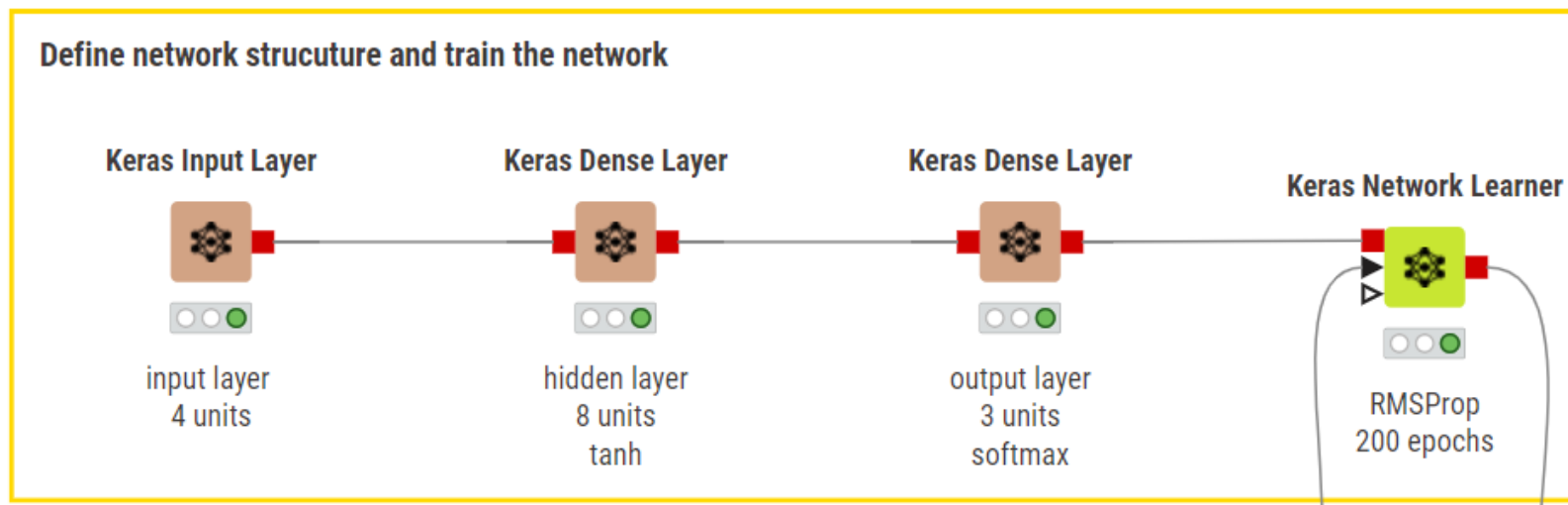
Info Nodes Explorer Monitor

1: Normalized table 2: Normalize Model Flow Variables

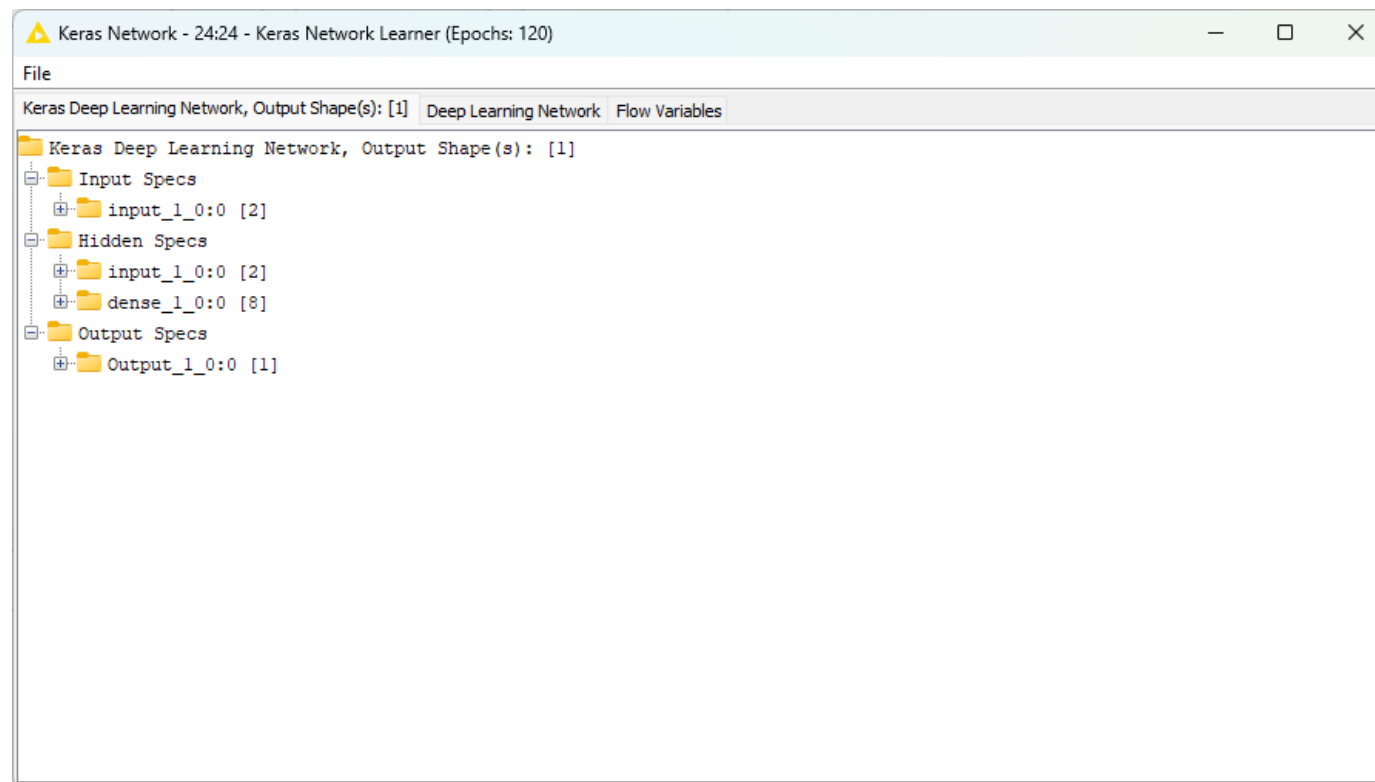
Rows: 112 | Columns: 6 Table Statistics

#	RowID	sepal_length Number (double)	sepal_width Number (double)	petal_length Number (double)	petal_width Number (double)	species Number (integer)	class_collection List
1	Row1	-1.152	-0.108	-1.288	-1.285	0	[0]
2	Row2	-1.395	0.35	-1.344	-1.285	0	[0]
3	Row3	-1.516	0.121	-1.233	-1.285	0	[0]
4	Row4	-1.031	1.266	-1.288	-1.285	0	[0]
5	Row5	-0.546	1.953	-1.121	-1.022	0	[0]
6	Row6	-1.516	0.808	-1.288	-1.154	0	[0]
7	Row7	-1.031	0.808	-1.233	-1.285	0	[0]
8	Row8	-1.758	-0.337	-1.288	-1.285	0	[0]
9	Row9	-1.152	0.121	-1.233	-1.417	0	[0]
10	Row10	-0.546	1.495	-1.233	-1.285	0	[0]
11	Row12	-1.273	-0.108	-1.288	-1.417	0	[0]
12	Row15	-0.182	3.098	-1.233	-1.022	0	[0]
13	Row16	-0.546	1.953	-1.344	-1.022	0	[0]
14	Row17	-0.91	1.037	-1.288	-1.154	0	[0]
15	Row18	-0.182	1.724	-1.121	-1.154	0	[0]
16	Row19	-0.91	1.724	-1.233	-1.154	0	[0]
17	Row20	-0.546	0.808	-1.121	-1.285	0	[0]

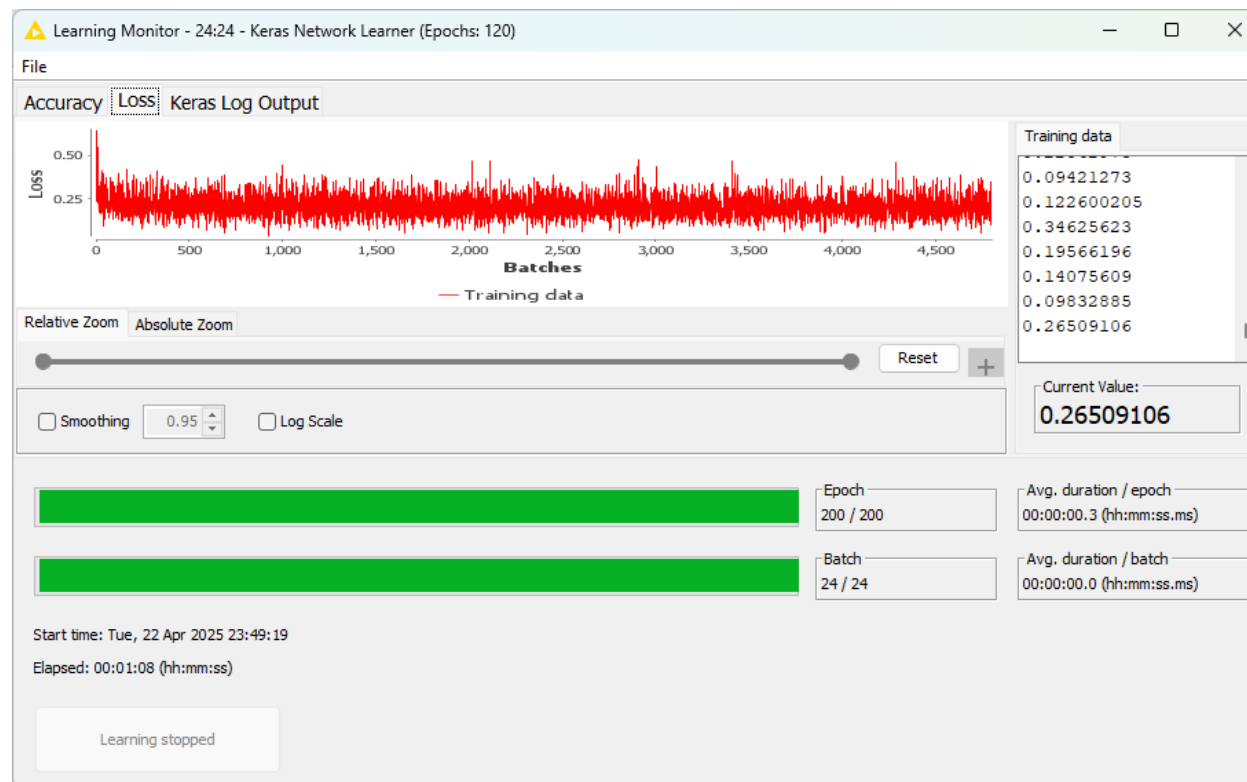
Part 2: Define network structure



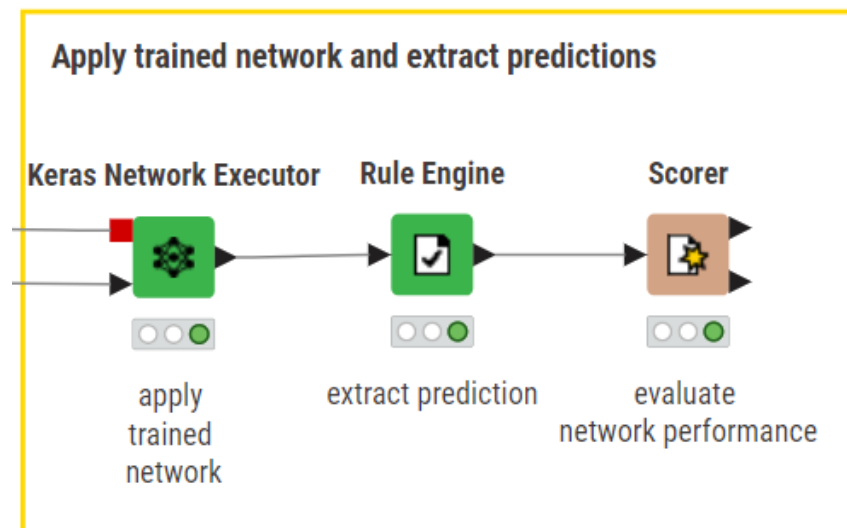
Part 2: Define network structure



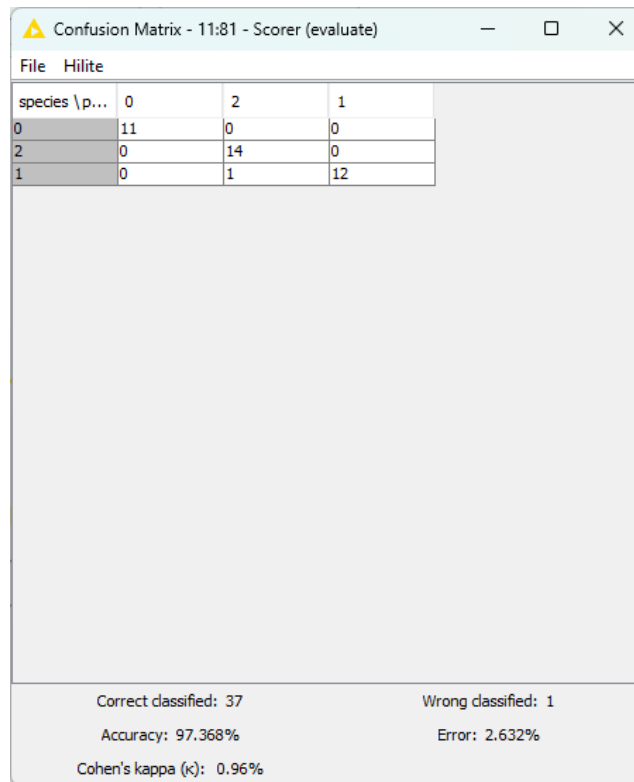
Part 2: Define network structure



Part 3: Apply trained network and extract predictions



Part 3: Apply trained network and extract predictions



A screenshot of a software window titled "Confusion Matrix - 11:81 - Scorer (evaluate)". The window contains a confusion matrix table and summary statistics at the bottom.

species \ p...	0	2	1
0	11	0	0
2	0	14	0
1	0	1	12

Summary statistics:

- Correct classified: 37
- Wrong classified: 1
- Accuracy: 97.368%
- Error: 2.632%
- Cohen's kappa (κ): 0.96%

Part 4: Compare performance with logistic regression

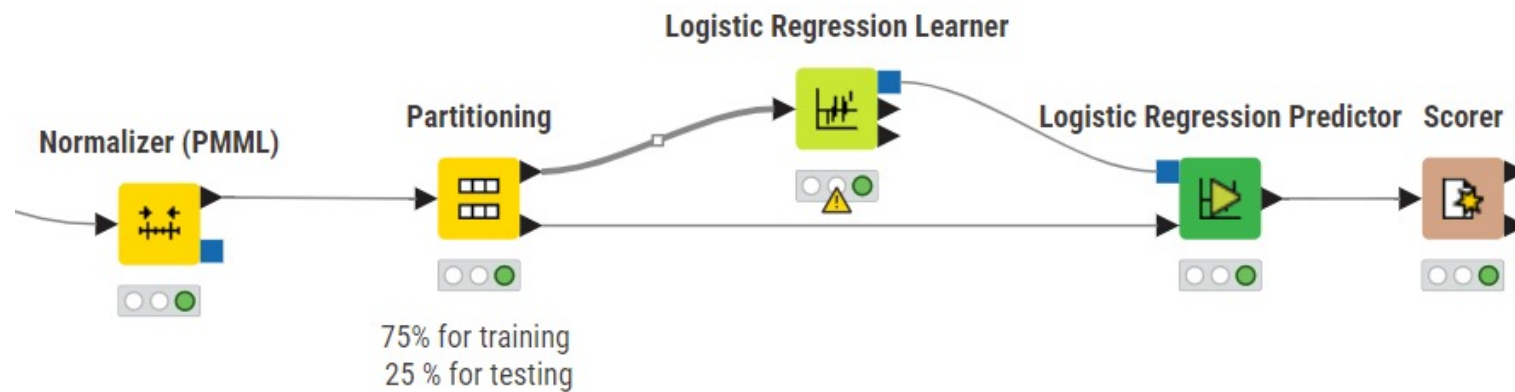
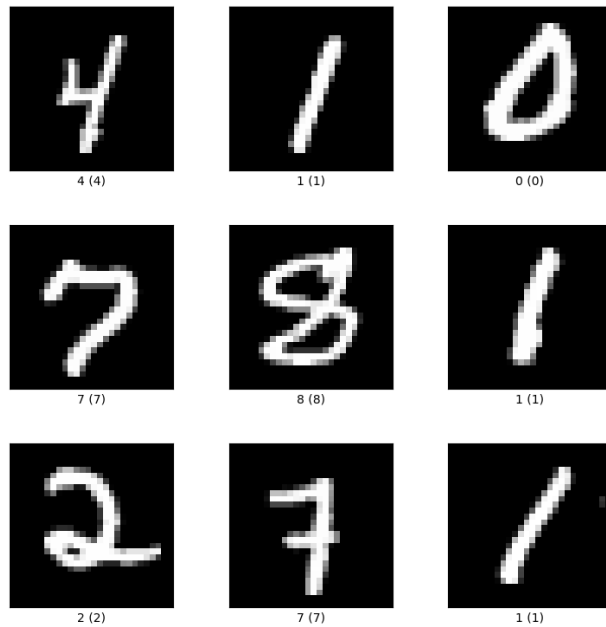


Image Classification:

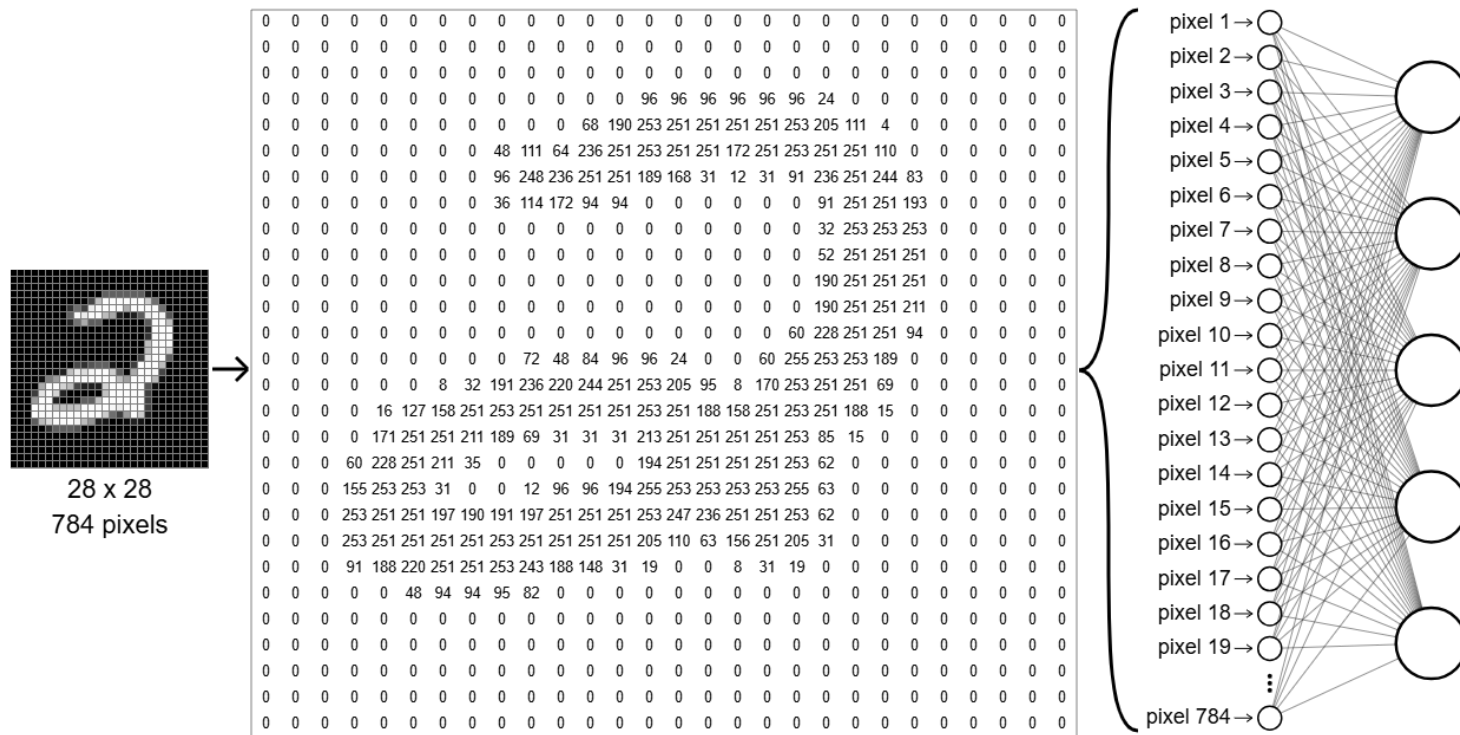
MNIST data



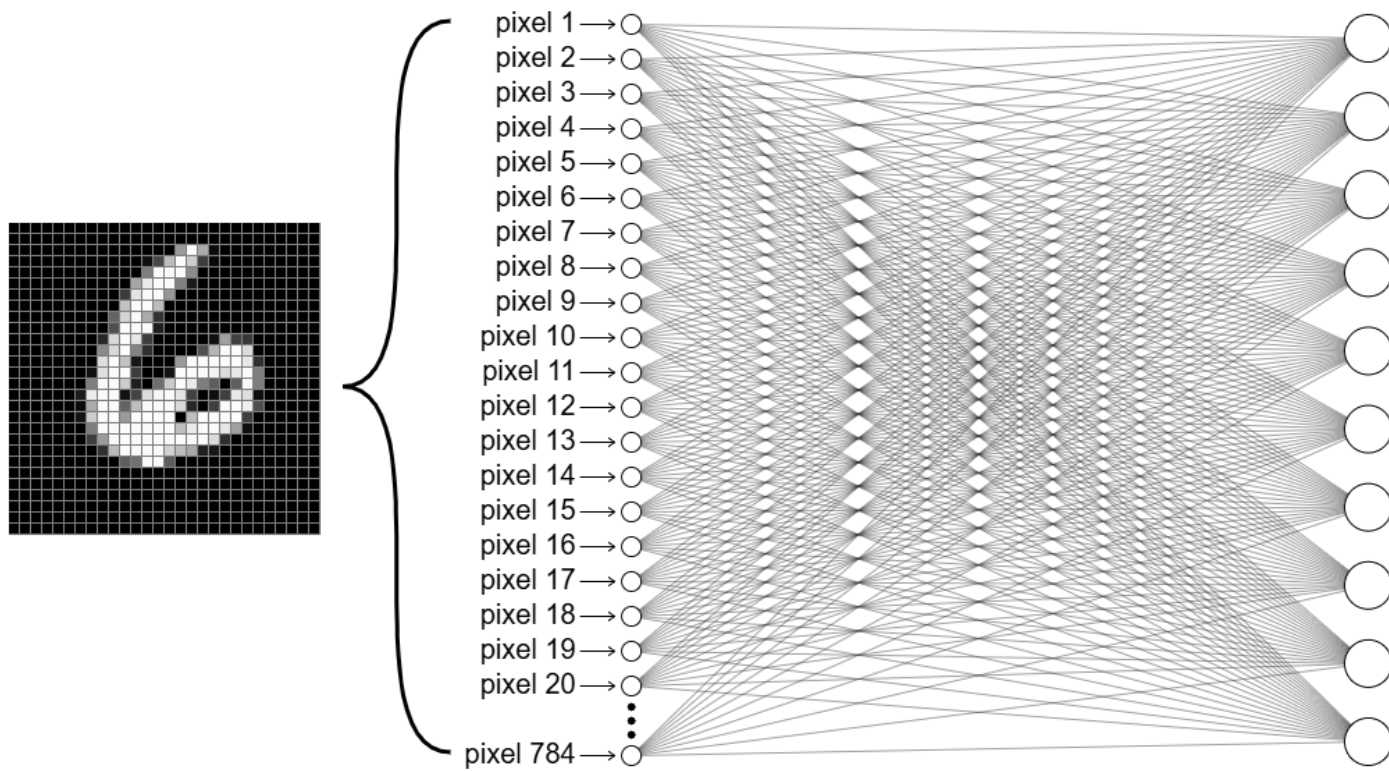
MNIST data



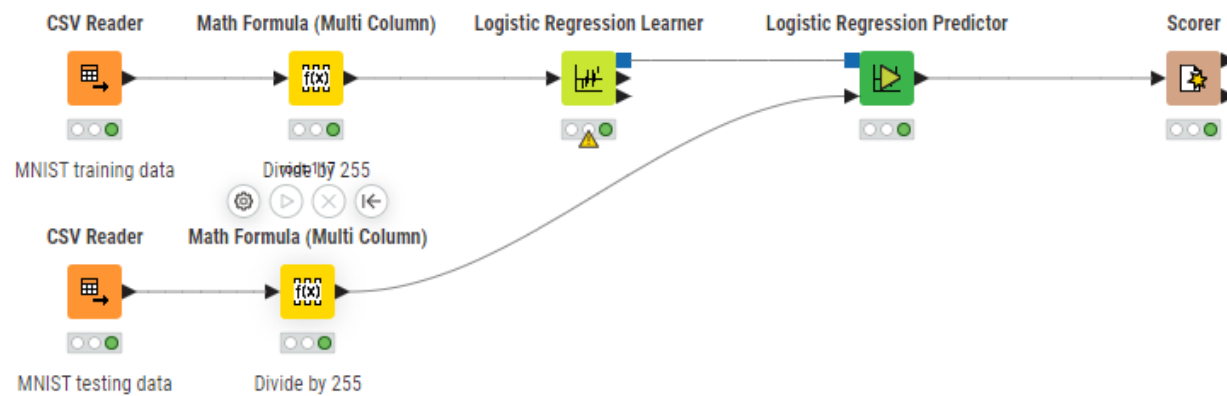
Data Representation



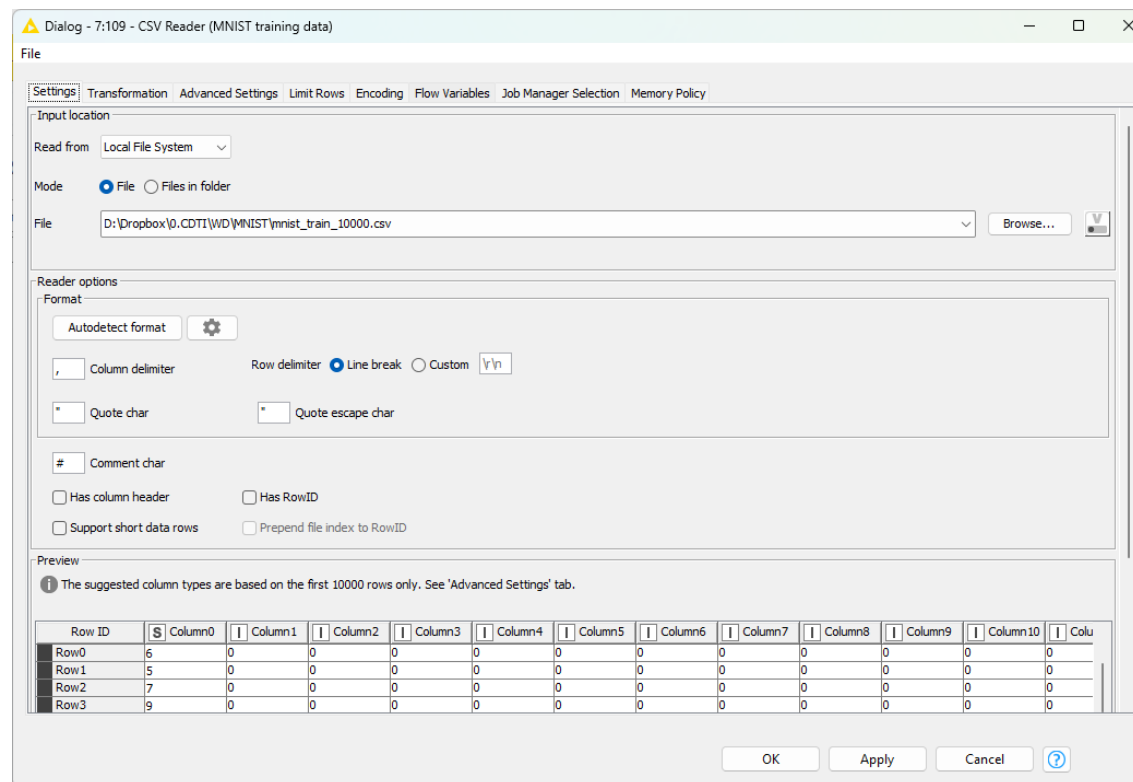
Logistic Regression



Logistic Regression



Important Setting: CSV reader training data



Important Setting: CSV reader testing data

Dialog - 7:113 - CSV Reader (MNIST testing data)

File

Settings Transformation Advanced Settings Limit Rows Encoding Flow Variables Job Manager Selection Memory Policy

Input location

Read from Local File System

Mode ☒ File ☐ Files in folder

File D:\Dropbox\0.CDTI\WD\MNIST\mnist_test_1500.csv Browse...

Reader options

Format

Autodetect format

Column delimiter , Row delimiter ☒ Line break ☐ Custom \r\n

Quote char " Quote escape char \"

Comment char

☐ Has column header ☐ Has RowID

☐ Support short data rows ☐ Prepend file index to RowID

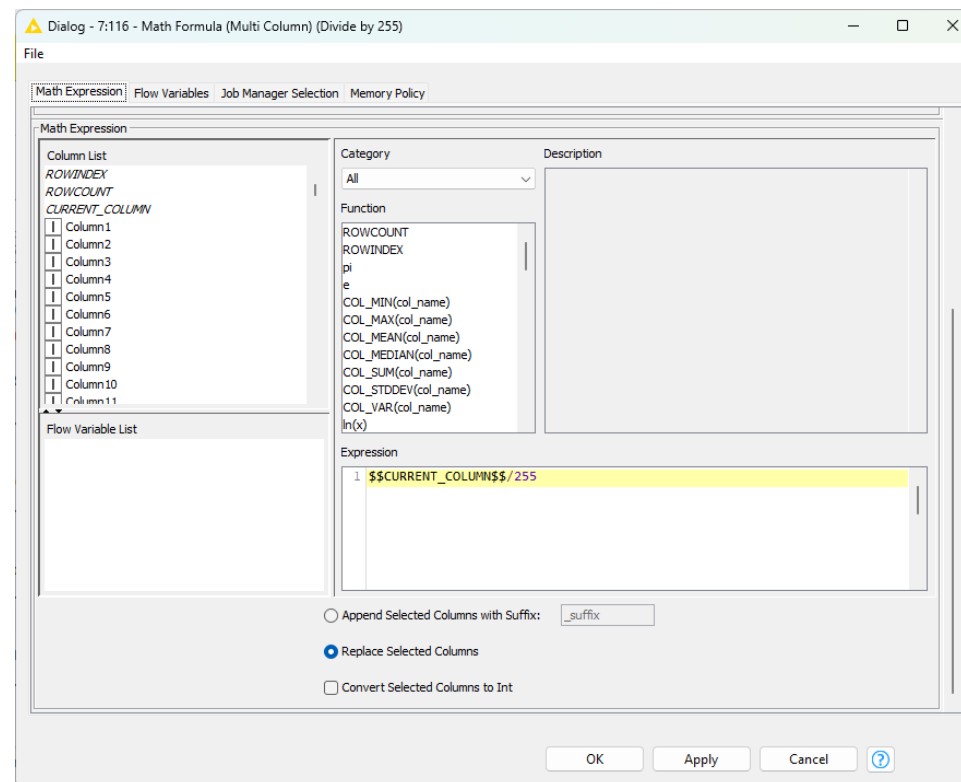
Preview

The suggested column types are based on the first 10000 rows only. See 'Advanced Settings' tab.

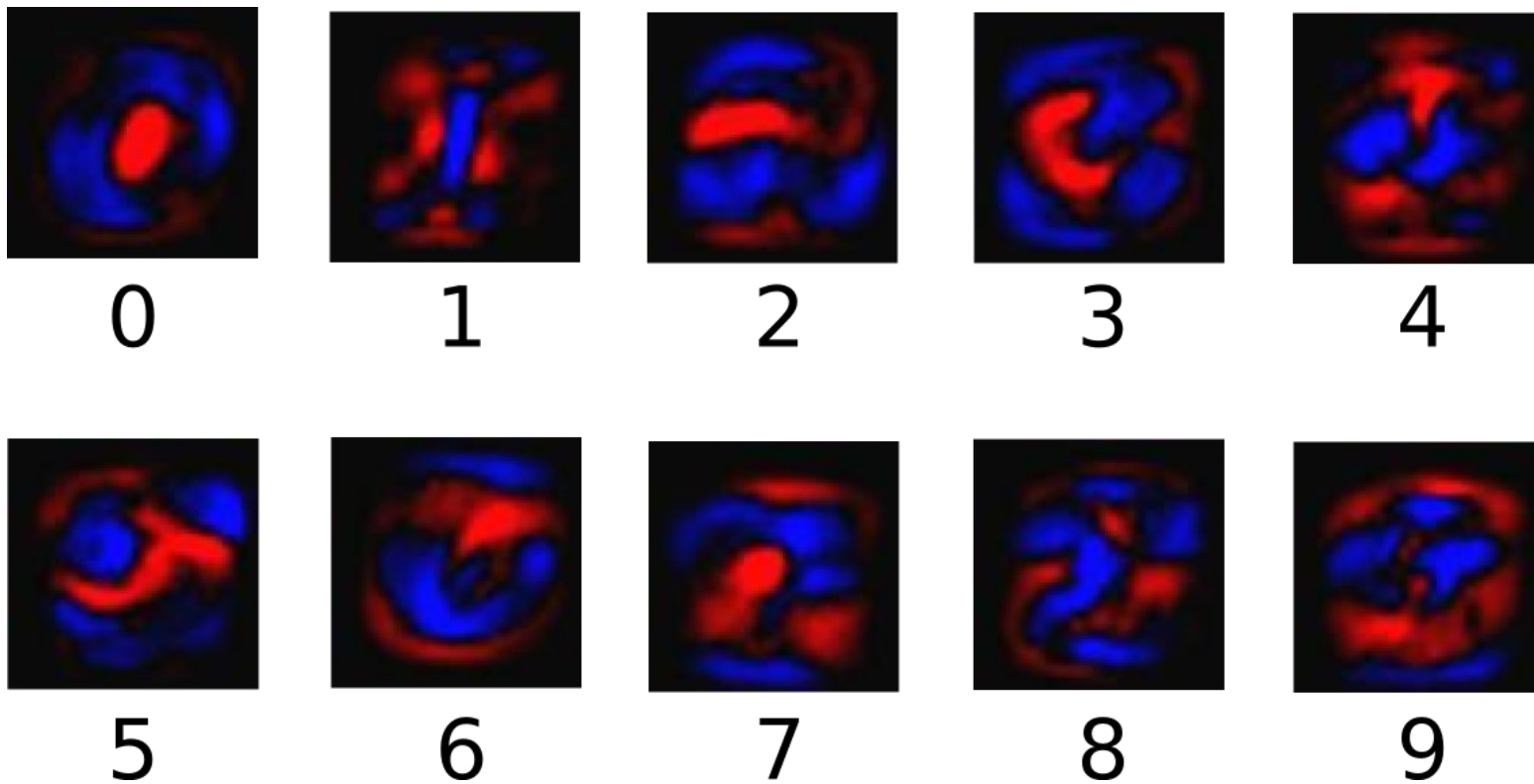
Row ID	S Column0	I Column1	I Column2	I Column3	I Column4	I Column5	I Column6	I Column7	I Column8	I Column9	I Column10	I Column11
Row0	7	0	0	0	0	0	0	0	0	0	0	0
Row1	2	0	0	0	0	0	0	0	0	0	0	0
Row2	1	0	0	0	0	0	0	0	0	0	0	0
Row3	0	0	0	0	0	0	0	0	0	0	0	0

OK Apply Cancel ?

Important Setting: Math Formula (Multi Column)



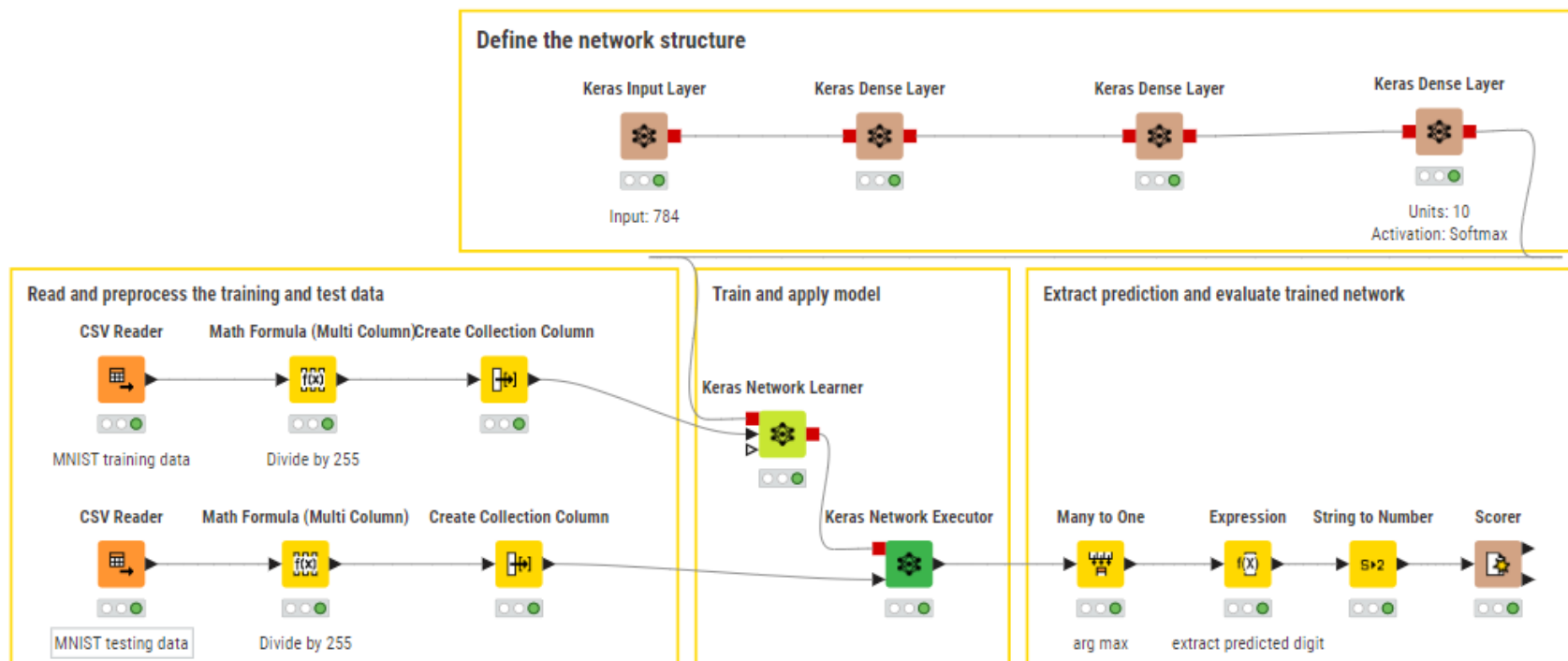
Learned Weights



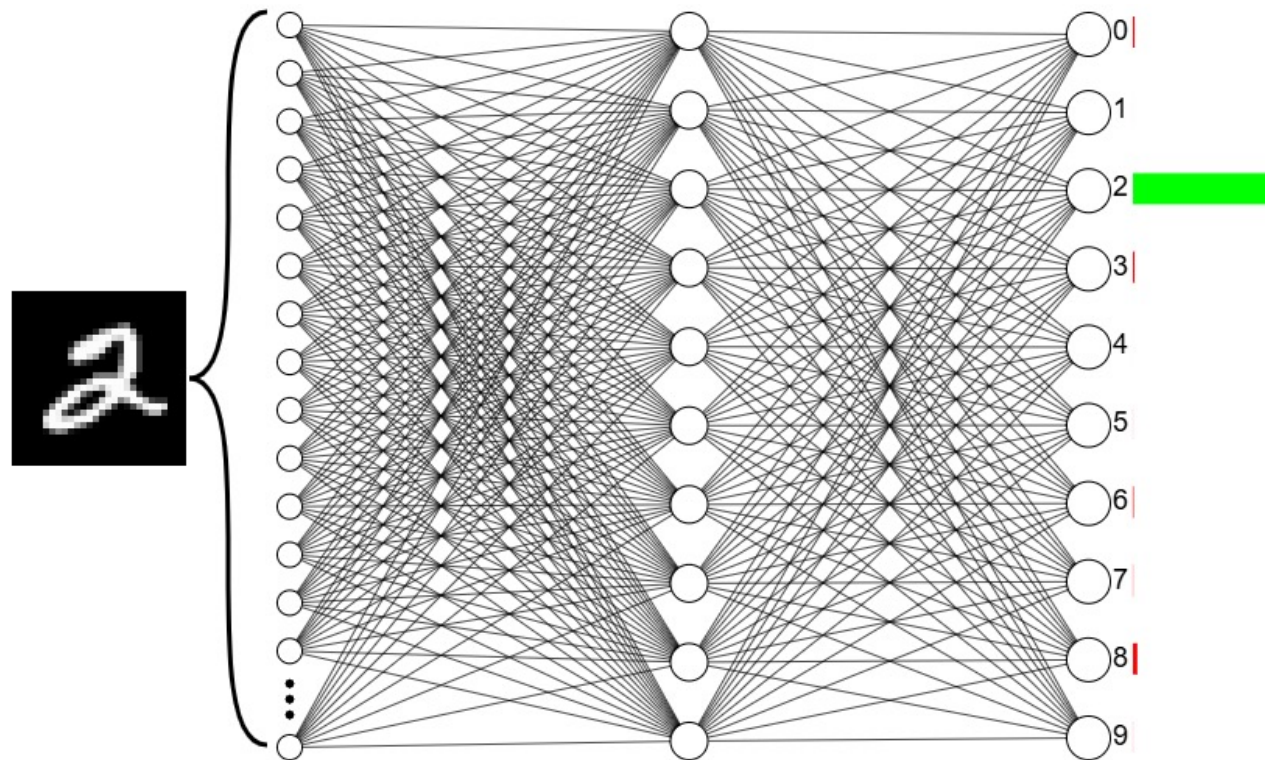
Multiclass classification: Logistic Regression

Confusion Matrix - 7:115 - Scorer										
File Hilite										
Column0 \ ...	7	2	1	0	4	9	5	6	3	8
7	125	6	1	1	1	10	1	0	3	2
2	4	139	2	2	2	1	3	6	6	7
1	1	0	170	0	0	1	1	1	2	3
0	1	2	0	115	0	0	4	3	1	0
4	3	2	0	0	140	11	0	5	5	1
9	4	1	1	1	10	117	1	0	5	4
5	1	2	0	2	1	1	102	6	13	9
6	1	4	0	3	0	0	5	117	1	0
3	5	2	4	0	1	4	11	1	122	4
8	2	8	1	0	4	6	10	2	9	98
Correct classified: 1,245										
Wrong classified: 255										
Accuracy: 83%										
Error: 17%										
Cohen's kappa (κ): 0.811%										

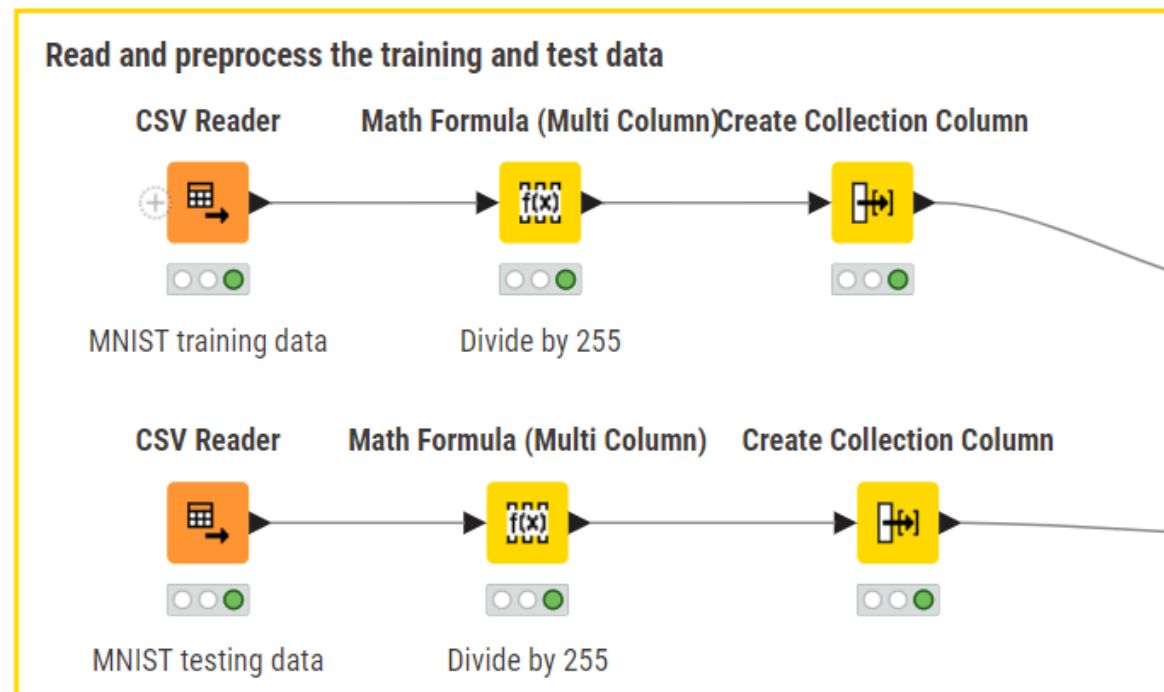
Multi Layer Perceptron (MLP)



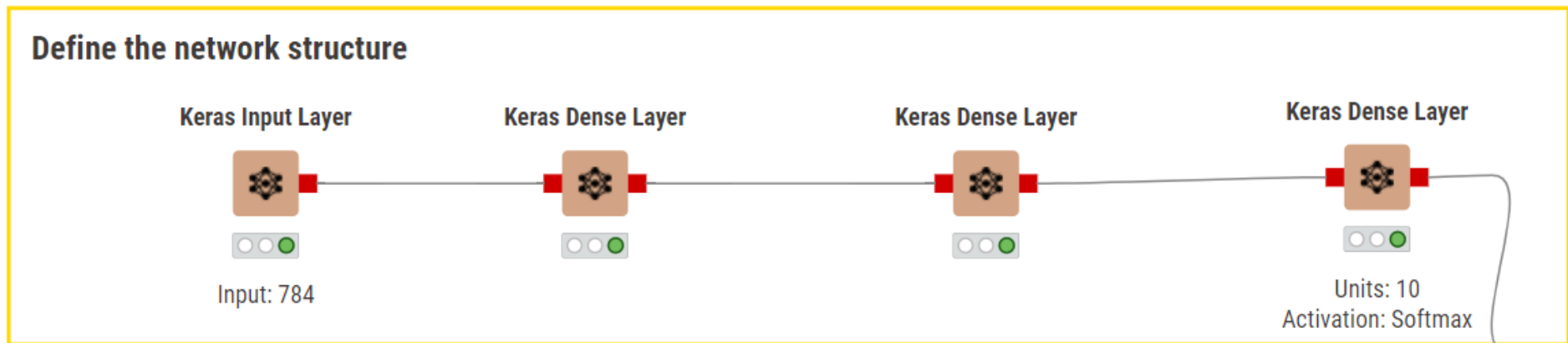
Multi Layer Perceptron (MLP)



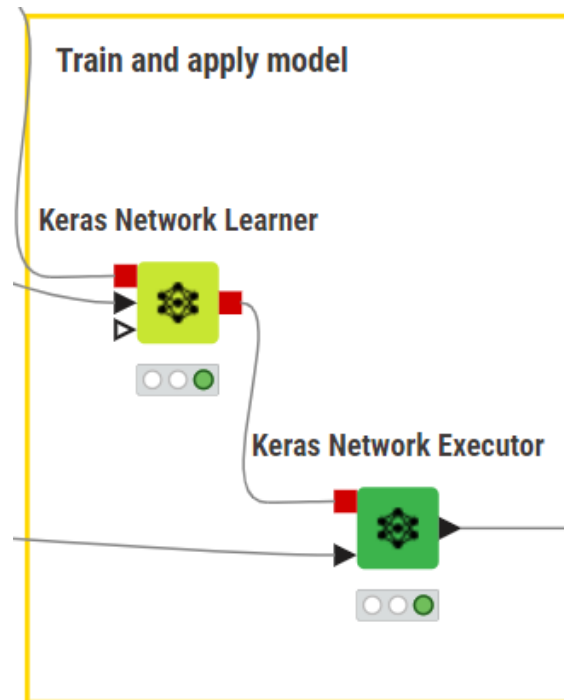
Part 1: Read and preprocess the training and test data



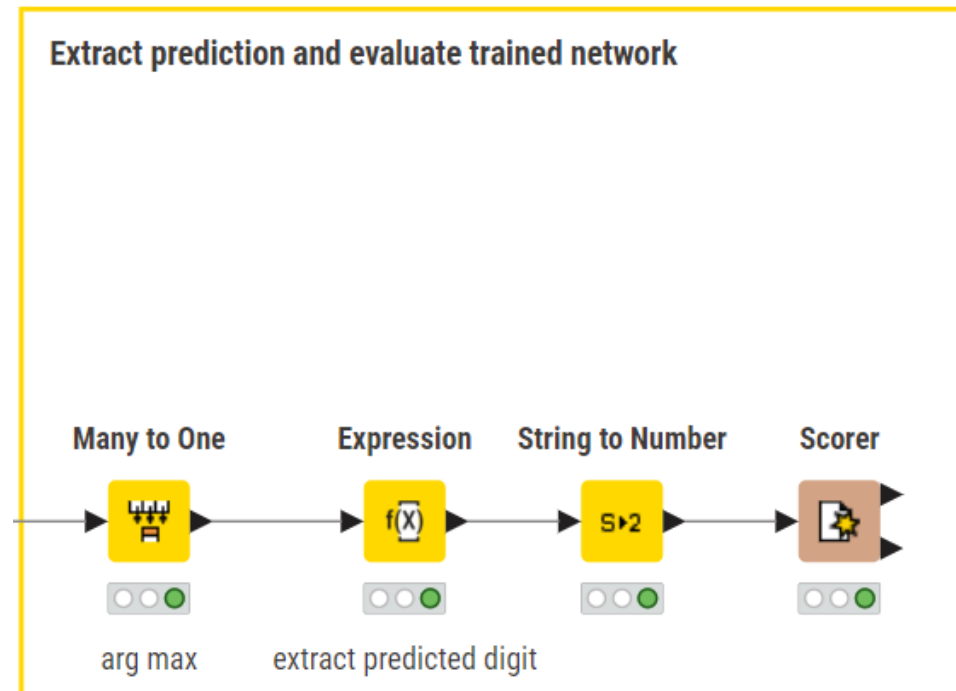
Part 2: Define the network structure



Part 3: Train and apply model



Part 4: Extract and evaluate trained network



Multi Layer Perceptron (MLP) Result


Confusion Matrix - 3:101 - Scorer

—
□
✕

File Hilite

Column0 \ ...	7	2	1	0	4	9	5	6	3	8
7	139	2	2	0	0	4	0	0	1	2
2	3	156	1	0	0	0	1	2	6	3
1	1	0	177	0	0	0	0	1	0	0
0	0	0	0	124	1	0	0	1	0	0
4	3	1	2	0	149	8	0	3	0	1
9	1	0	0	1	5	130	0	0	2	5
5	0	0	0	5	0	1	113	2	11	5
6	0	2	0	4	0	0	4	121	0	0
3	3	4	1	1	0	1	3	0	139	2
8	1	5	3	2	5	2	8	0	1	113

Correct classified: 1,361

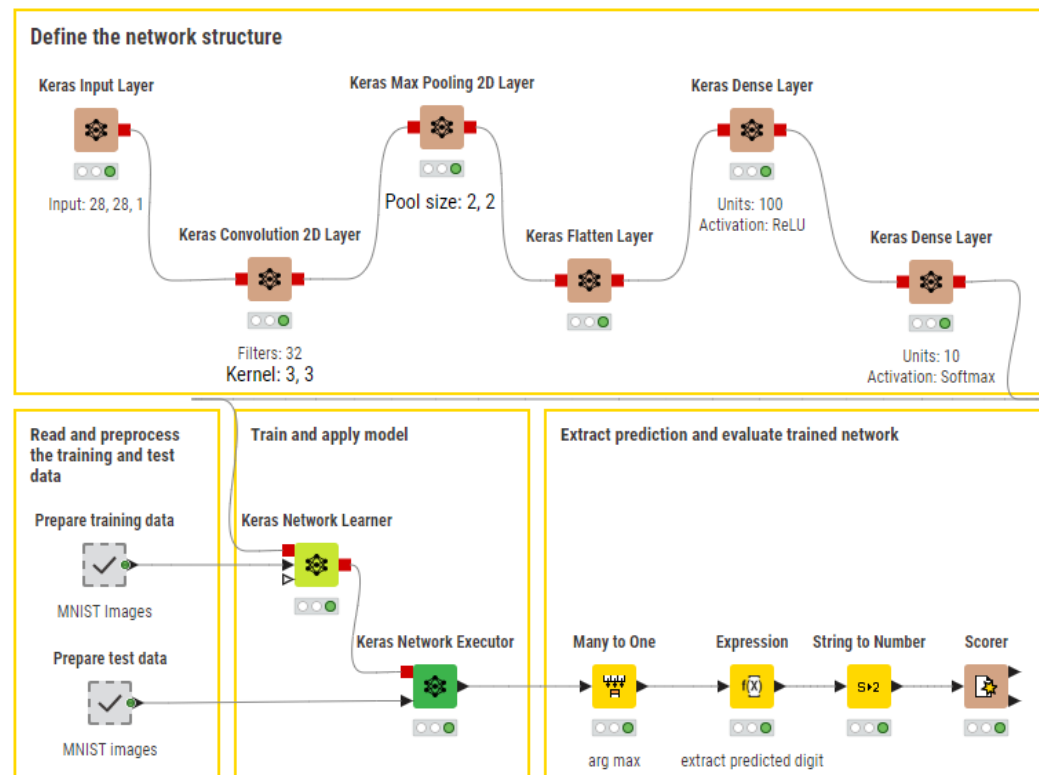
Wrong classified: 139

Accuracy: 90.733%

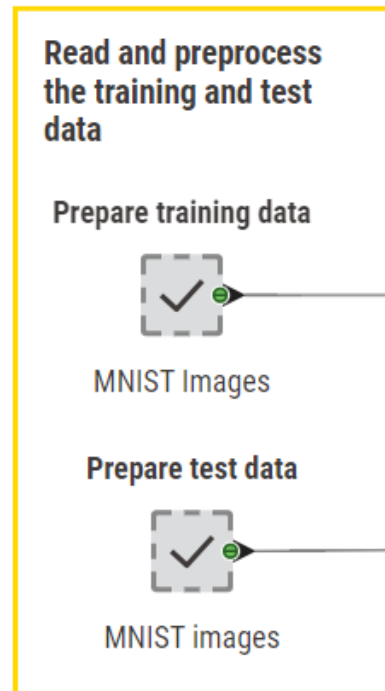
Error: 9.267%

Cohen's kappa (κ): 0.897%

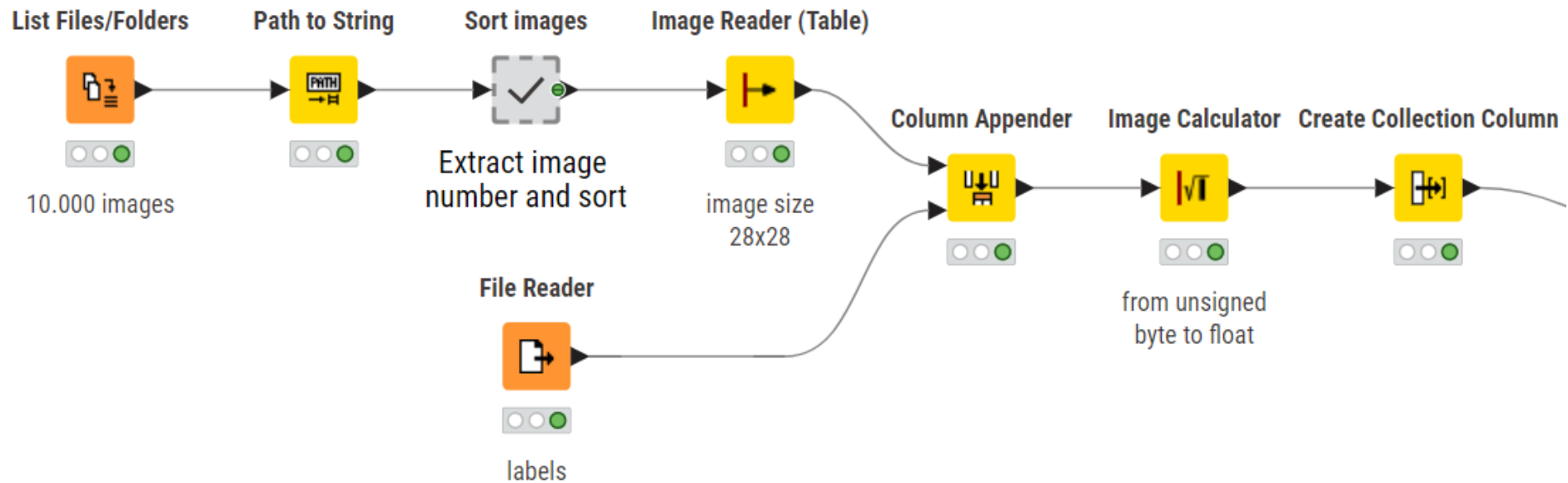
Convolution Neural Networks (CNN)



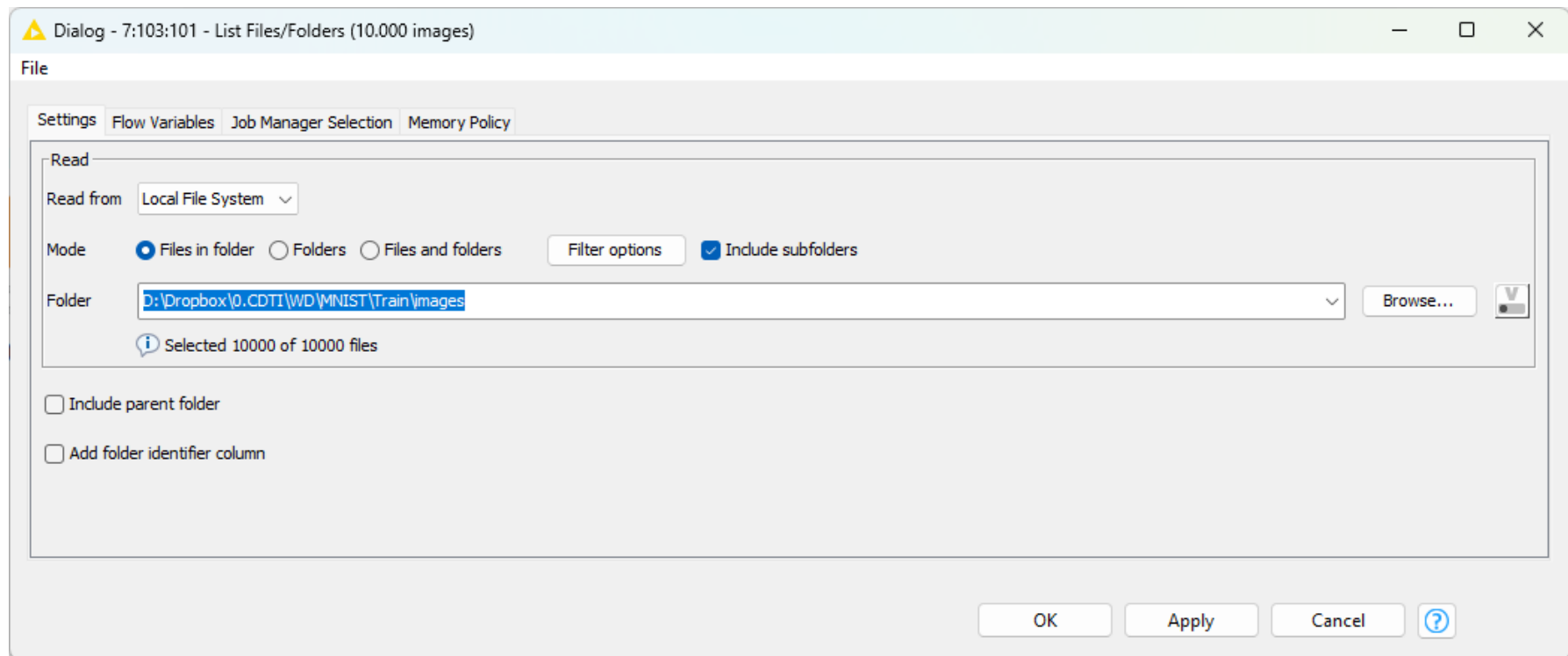
Part 1: Read and preprocess the training and test data



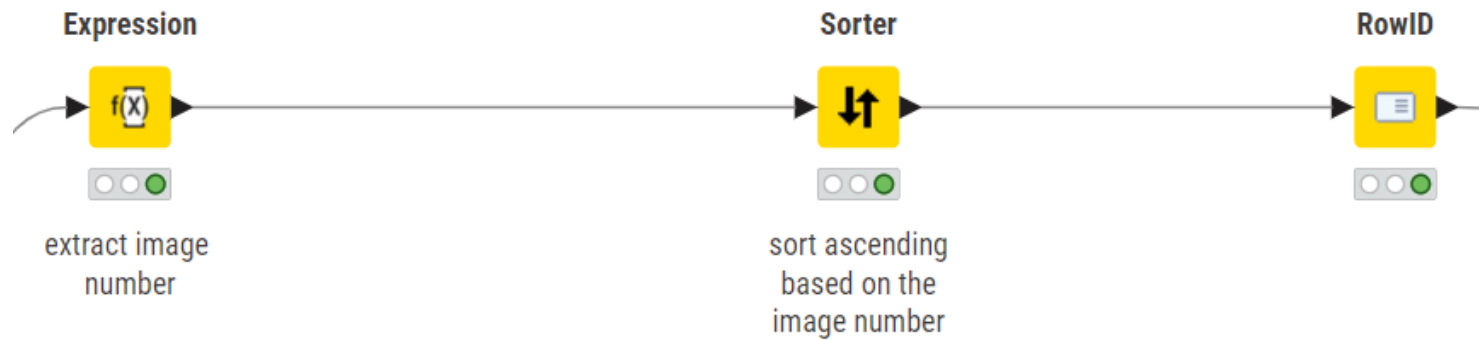
Preprocess training data



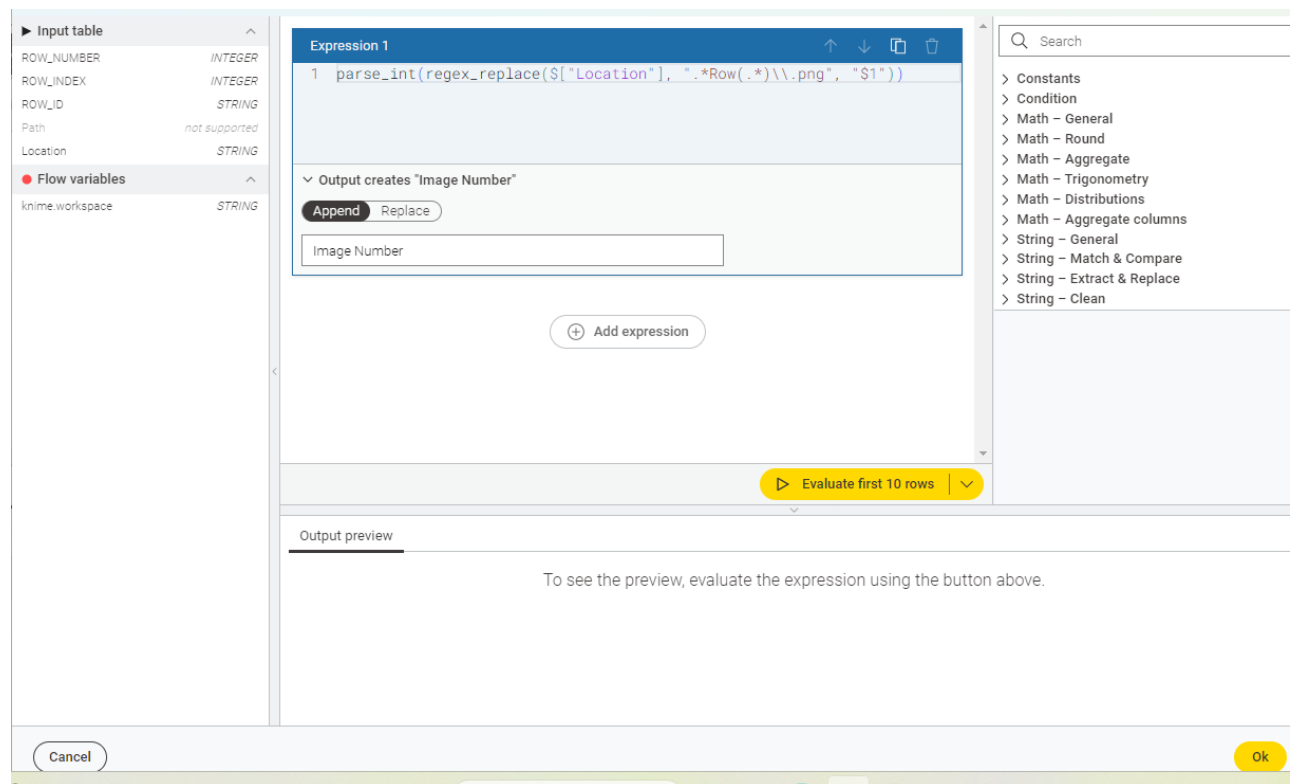
Important Setting: List Files/Folders



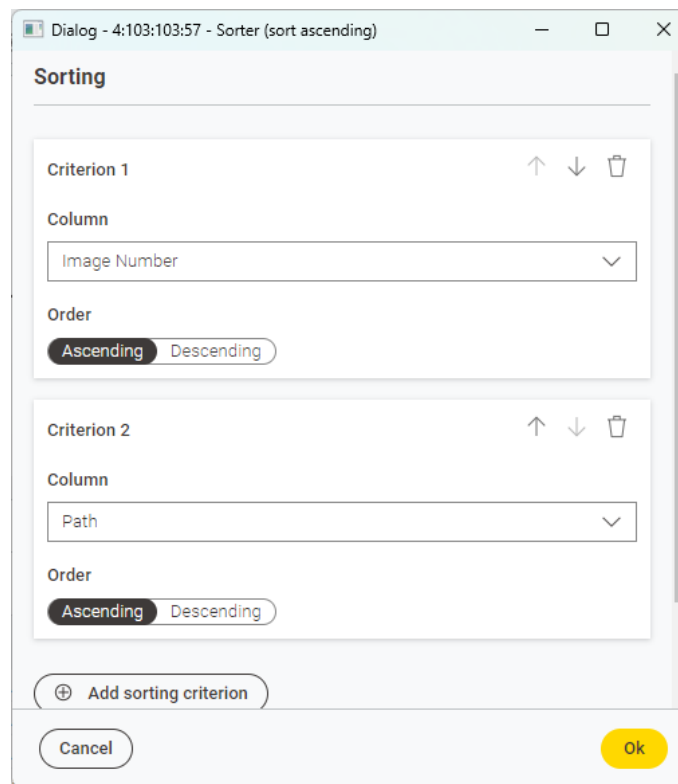
Sort Image



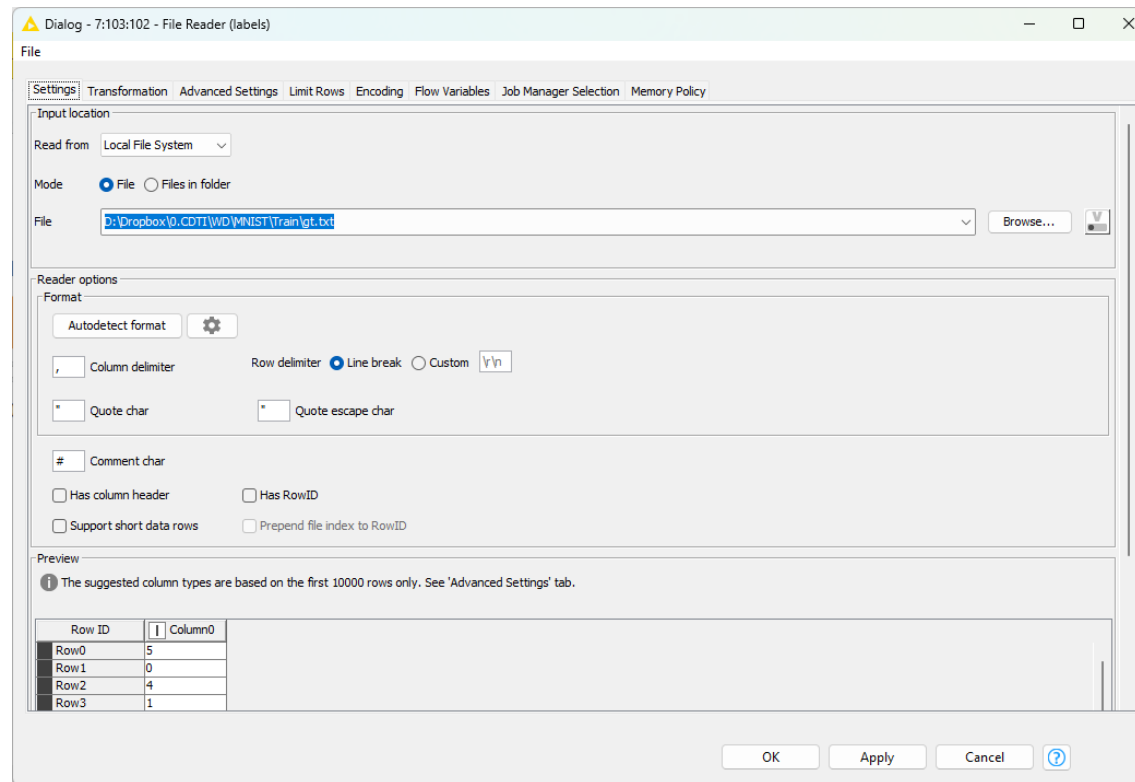
Important Setting: Expression



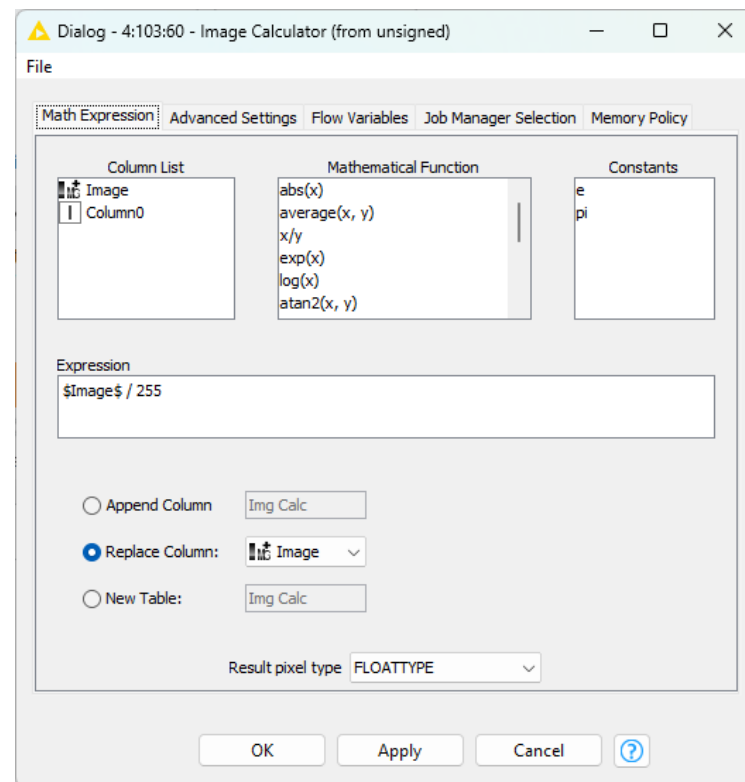
Important Setting: Sorter



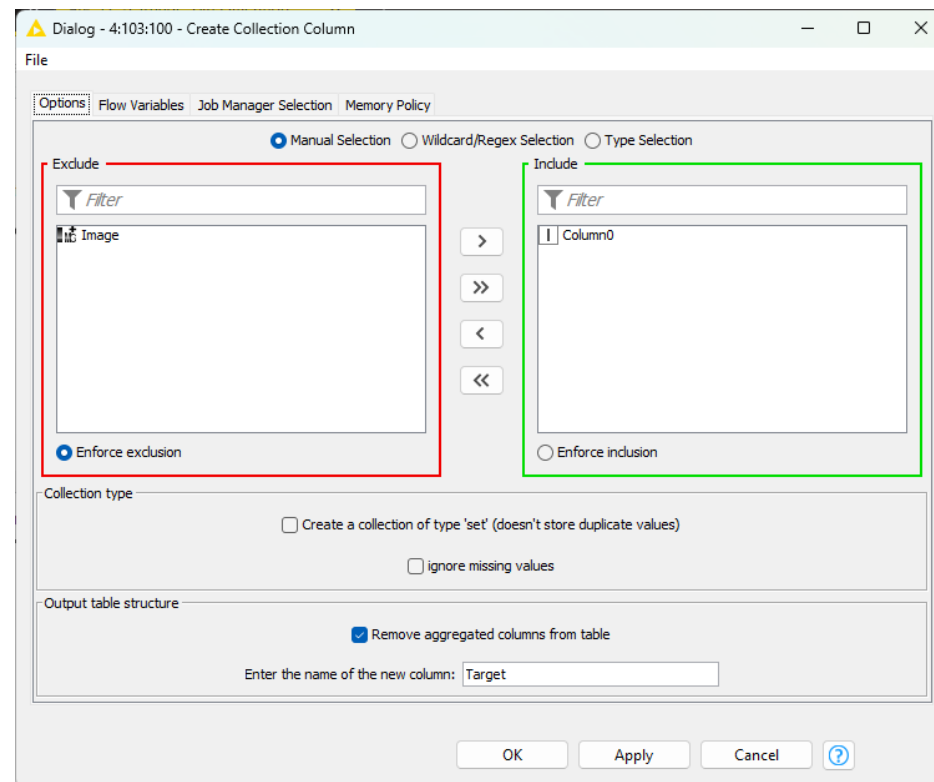
Important Setting: File Reader



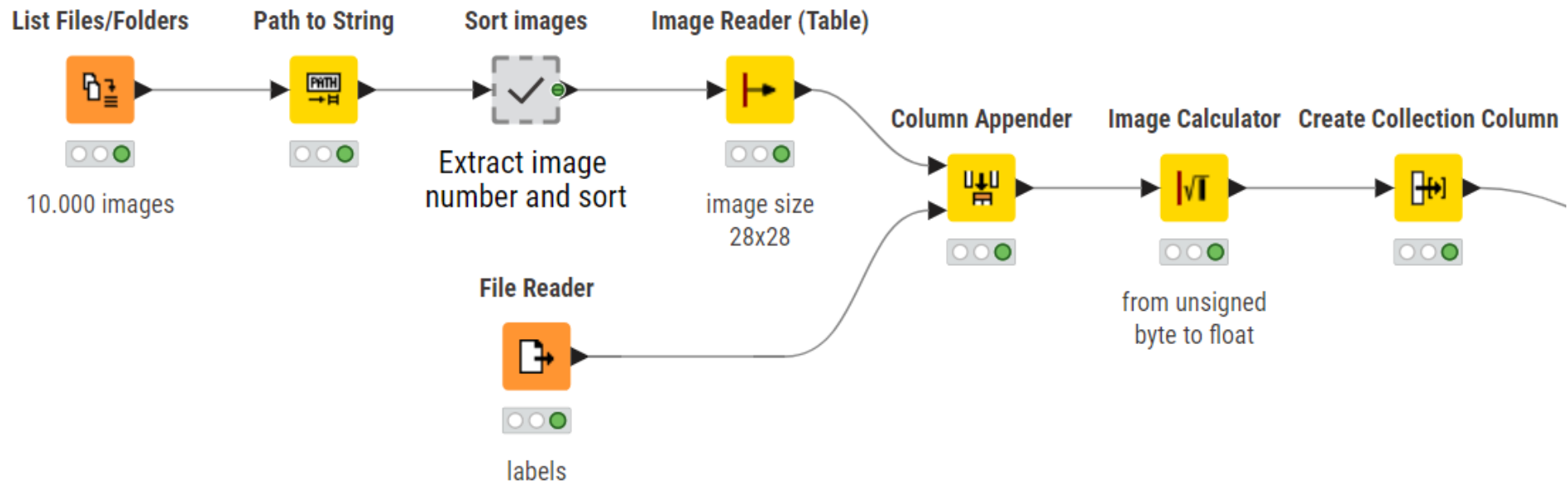
Important Setting: Image Calculator



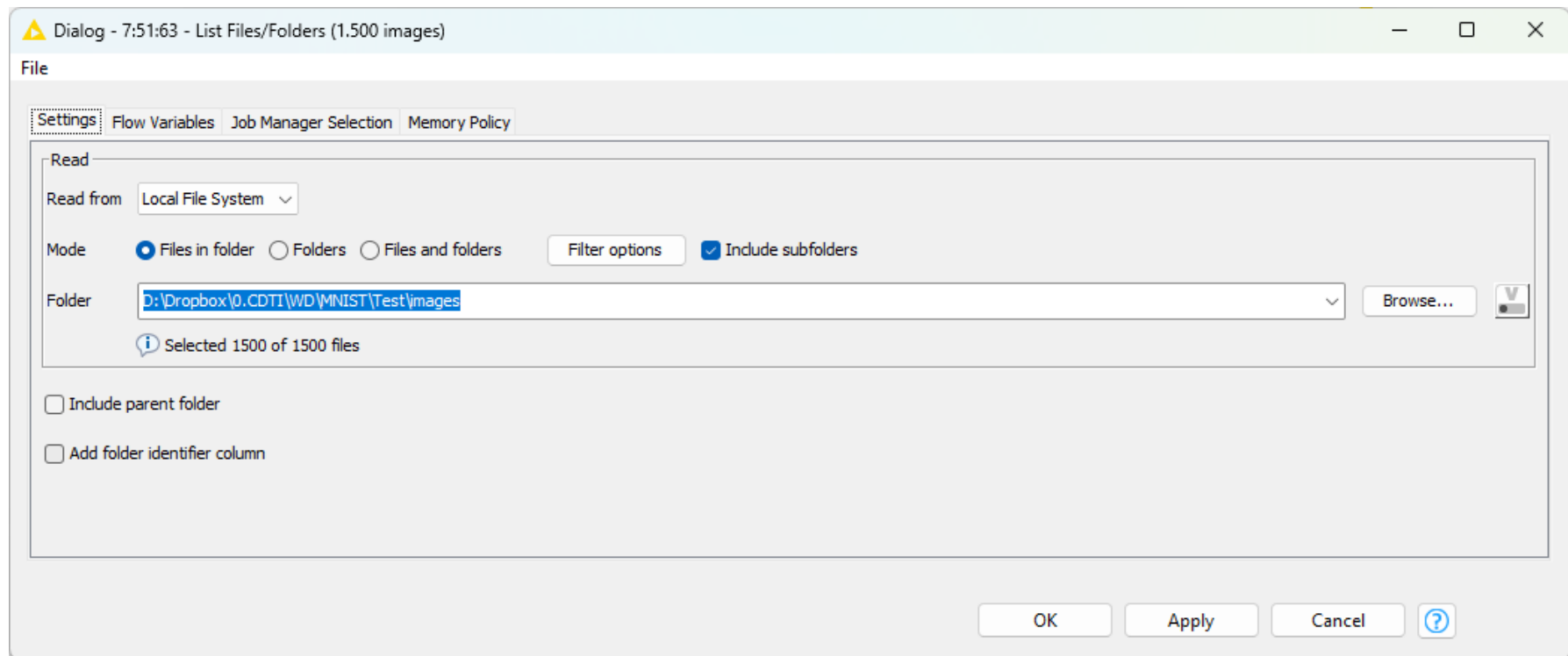
Important Setting: Create Collection Column



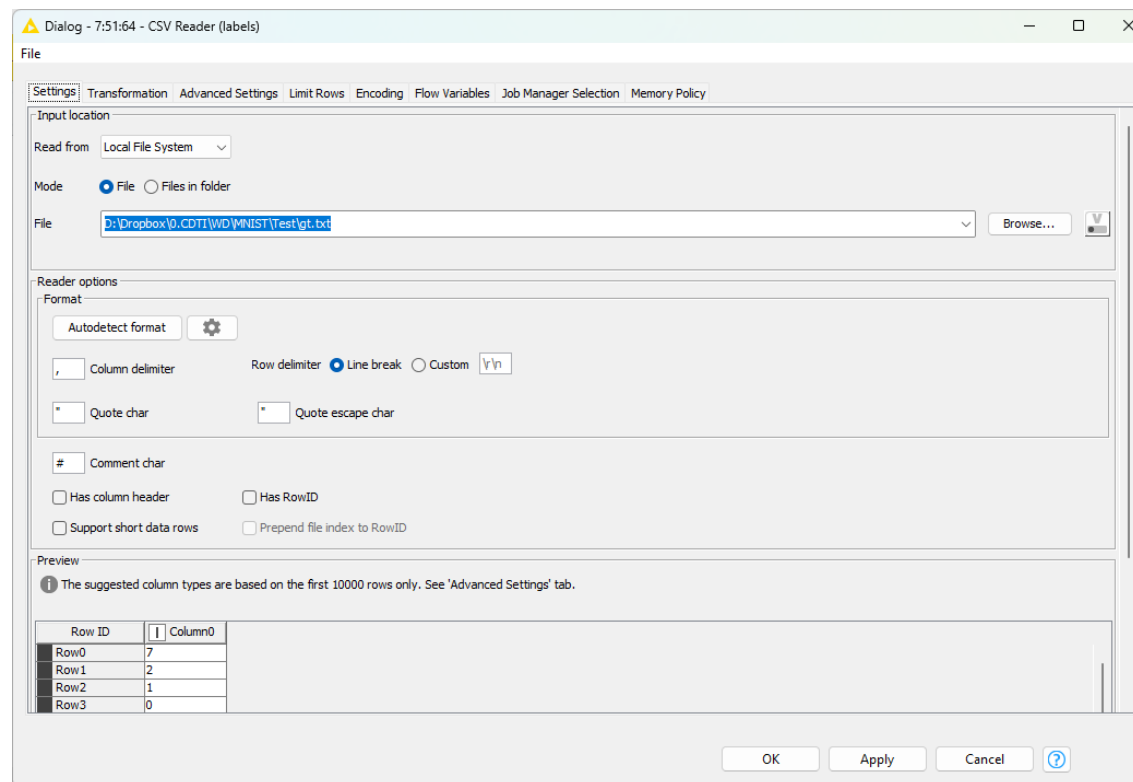
Preprocess testing data



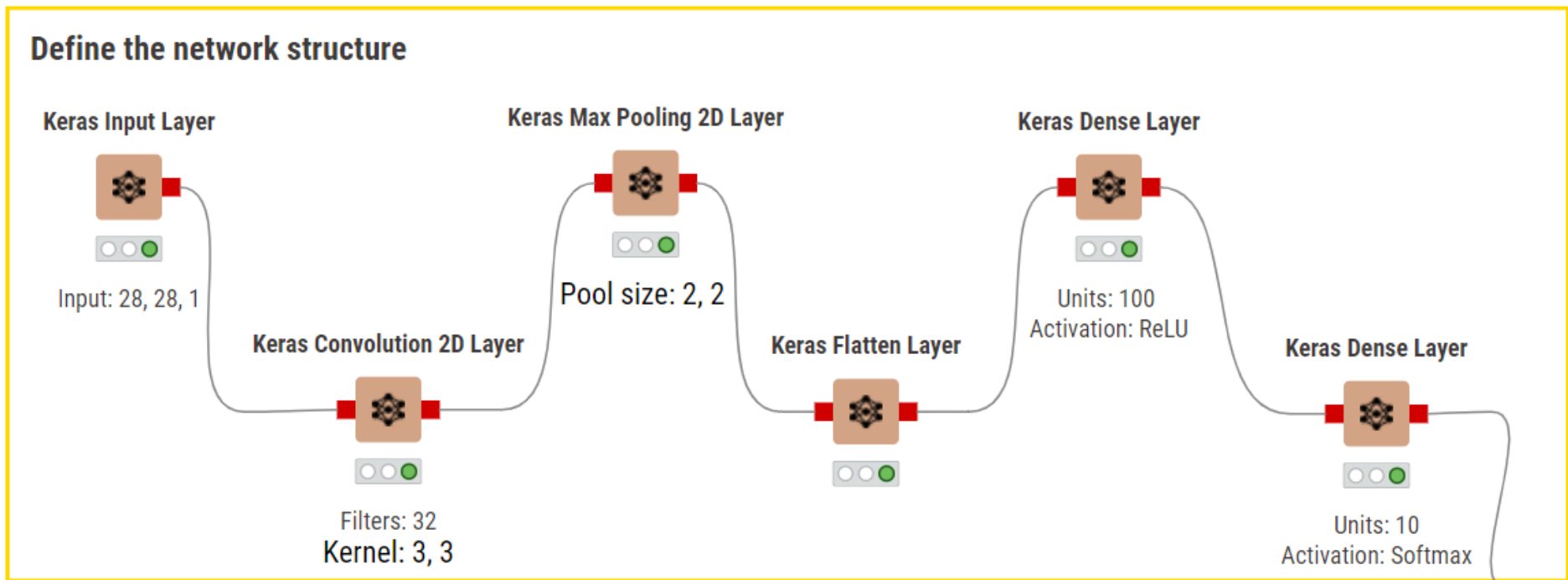
Important Setting: List Files/Folders



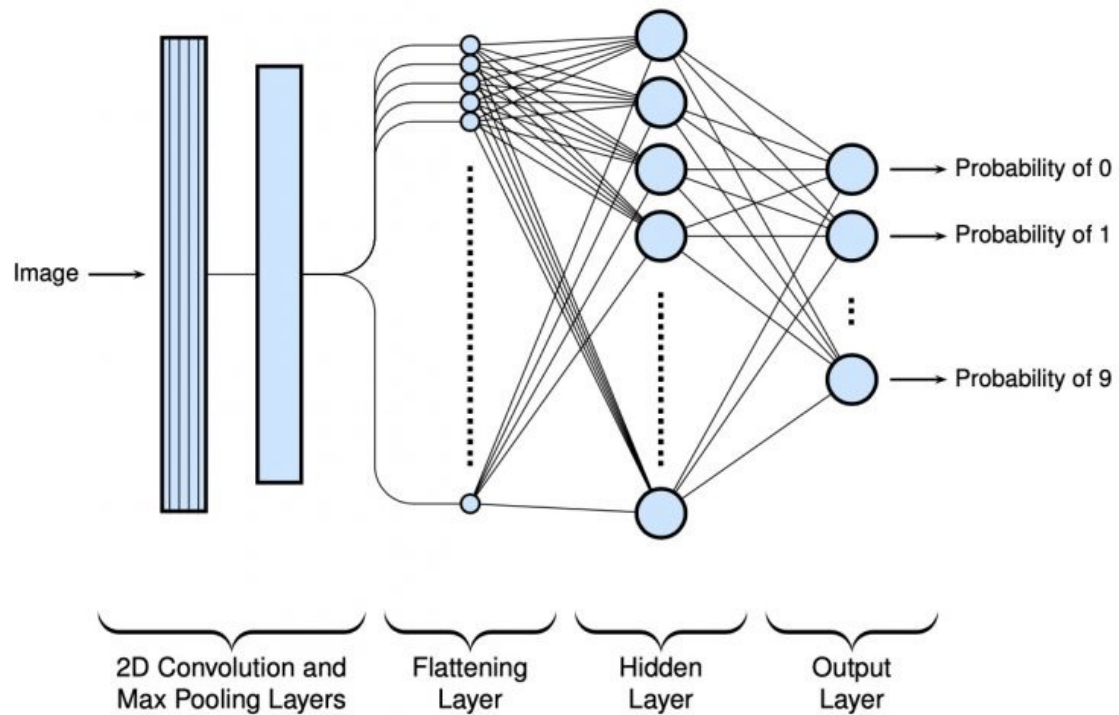
Important Setting: File Reader



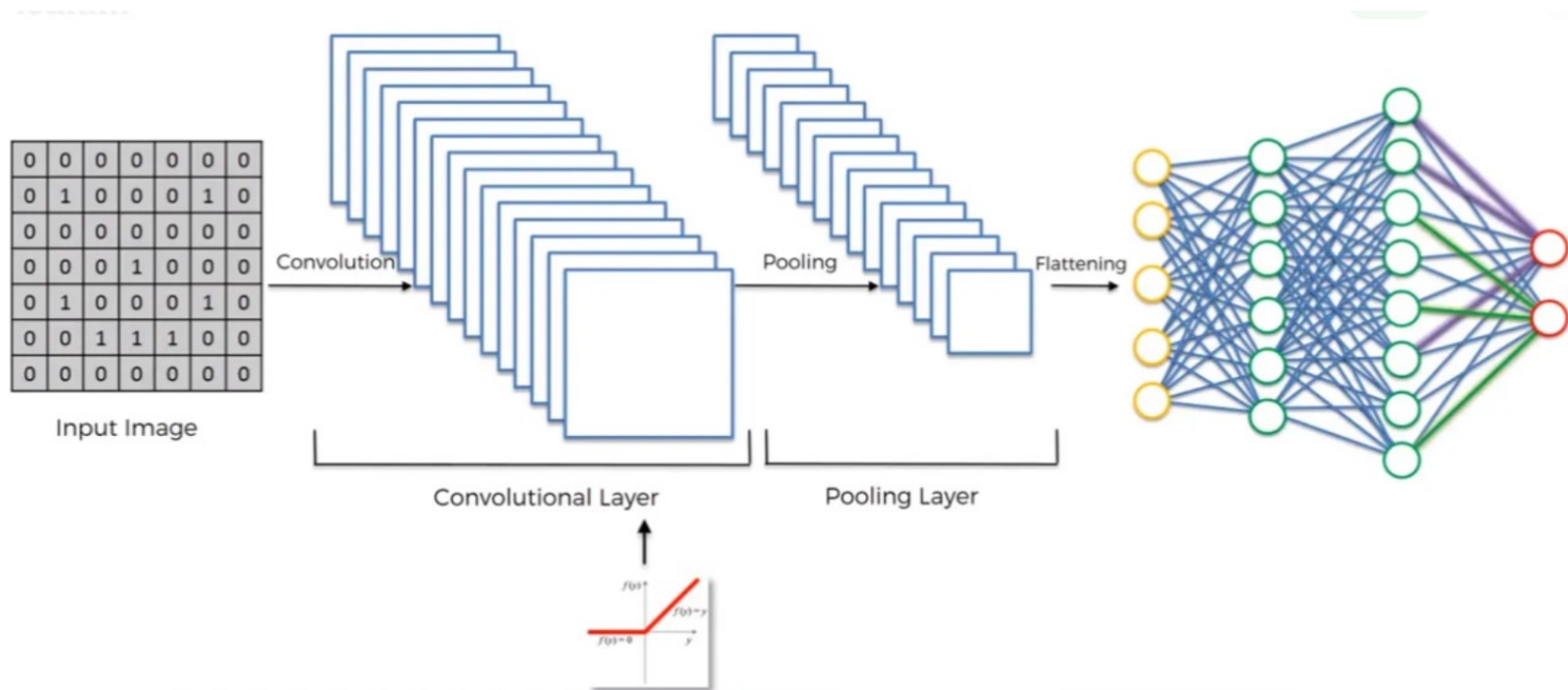
Part 2: Define the network structure



Part 2: Define the network structure

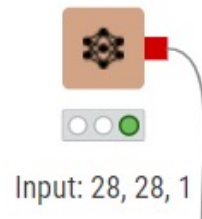


Part 2: Define the network structure



Important Setting: Keras Input Layer

Keras Input Layer



Dialog - 7:90 - Keras Input Layer (Input: 28, 28, 1)

File

Options | Flow Variables | Job Manager Selection

Name prefix ☐

Shape

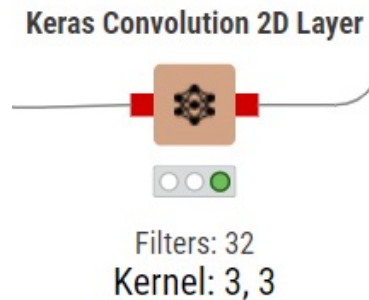
Batch size ☐

Data type

Data format

OK Apply Cancel ?

Important Setting: Keras Convolution 2D Layer



Dialog - 7:91 - Keras Convolution 2D Layer (Filters: 32)

File

Options Advanced Flow Variables Job Manager Selection

Name prefix ☐

Input tensor input_1_0:0 [28, 28, 1] float

Filters 32

Kernel size 3, 3

Strides 1, 1

Padding Valid

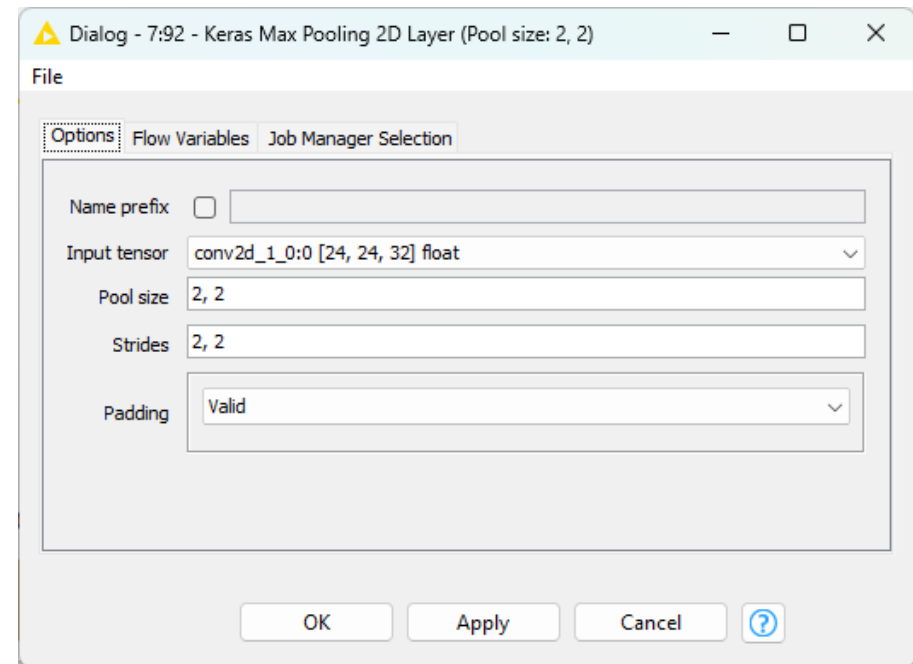
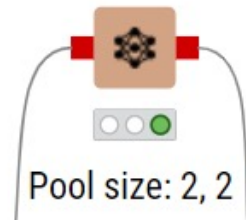
Dilation rate 2, 2

Activation function ReLU

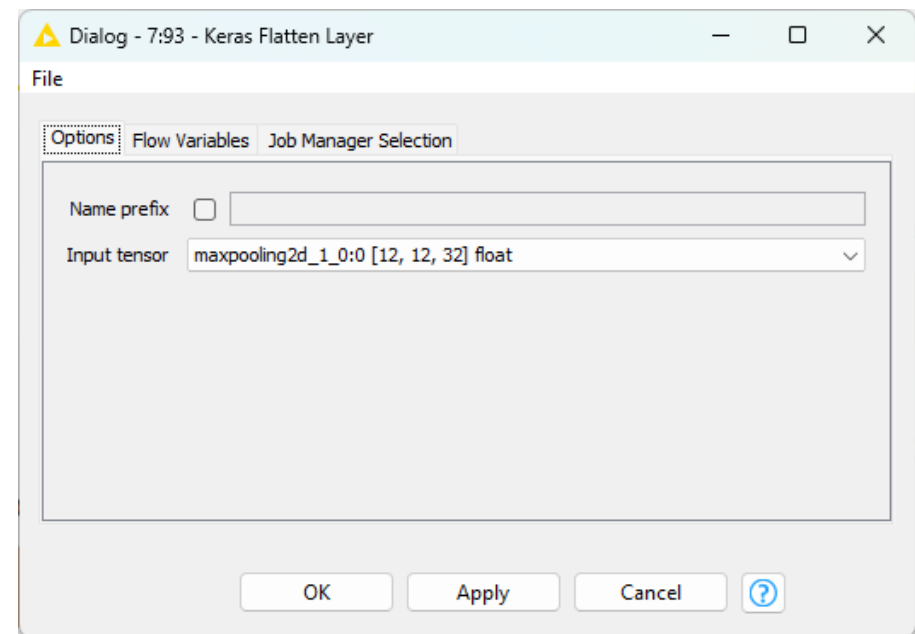
OK Apply Cancel ?

Important Setting: Keras Max Pooling 2D Layer

Keras Max Pooling 2D Layer

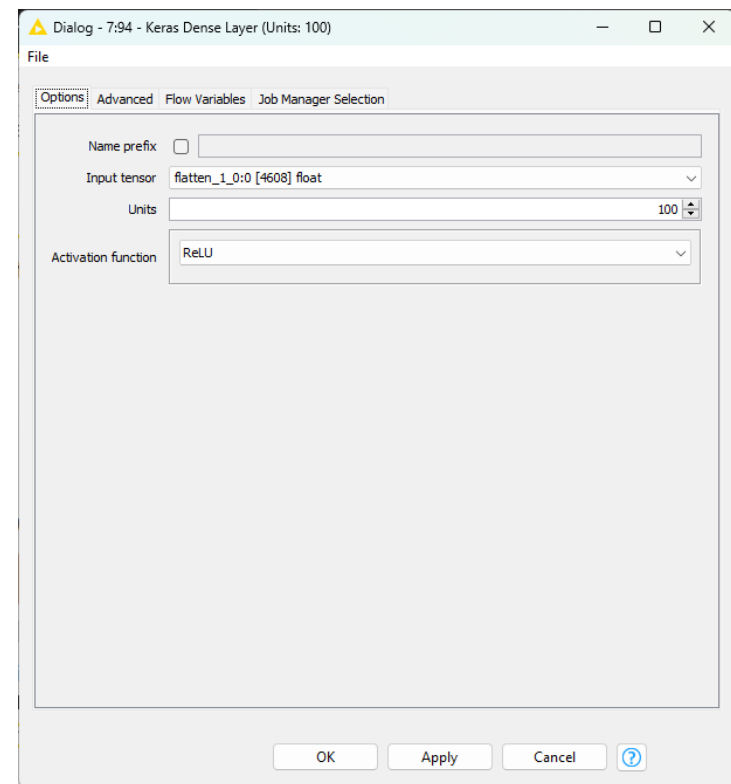
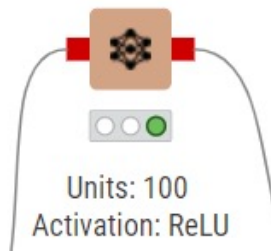


Important Setting: Keras Flatten Layer

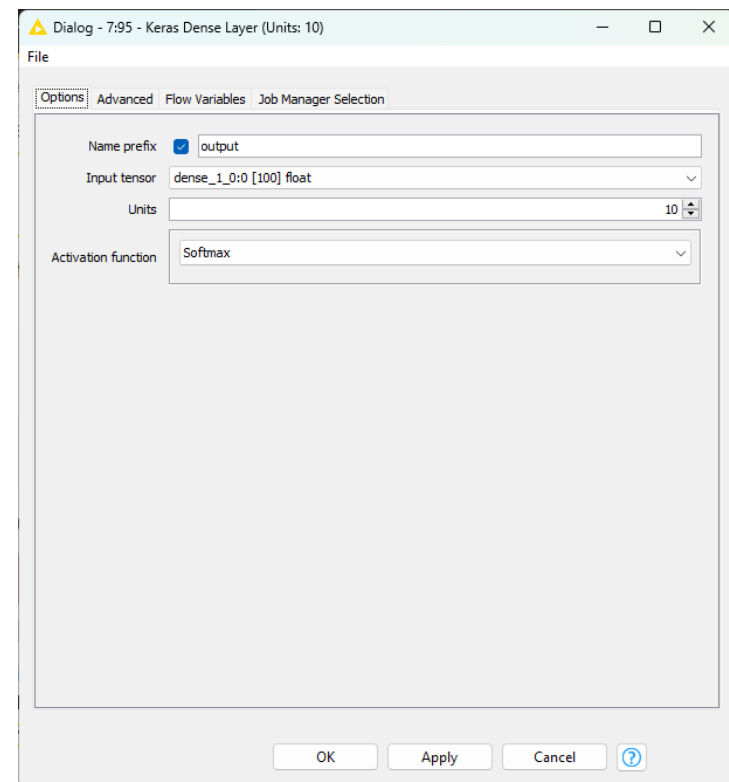
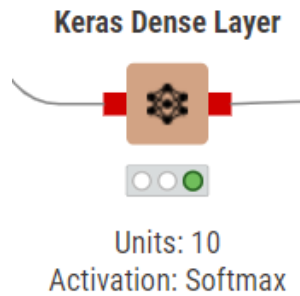


Important Setting: Keras Dense Layer

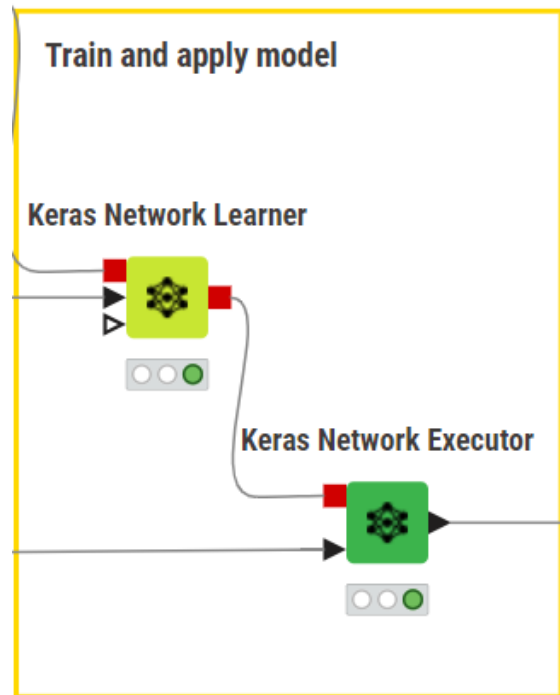
Keras Dense Layer



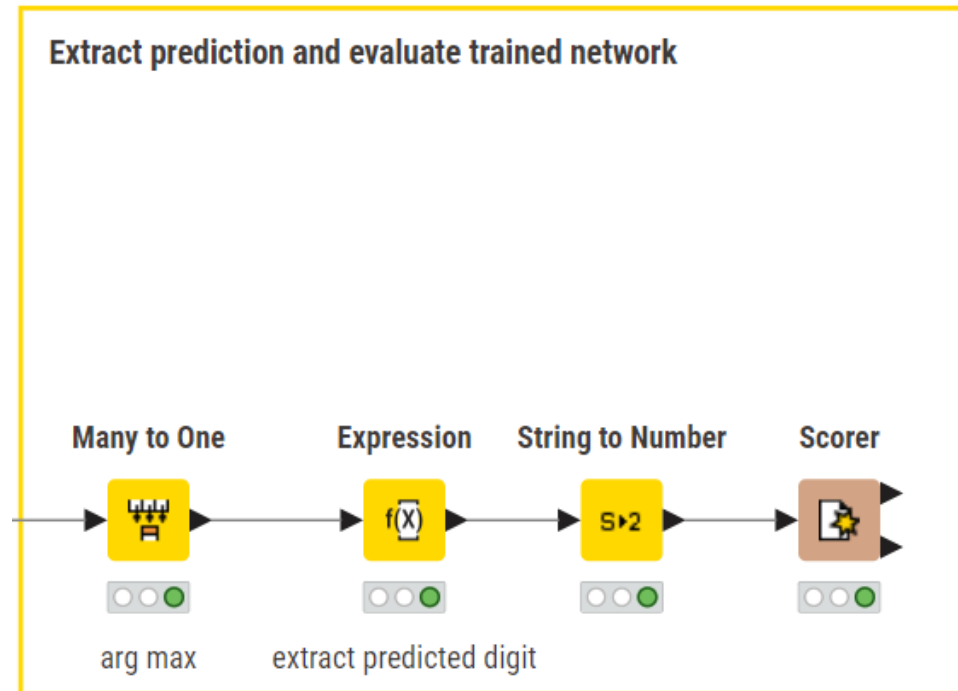
Important Setting: Keras Dense Layer



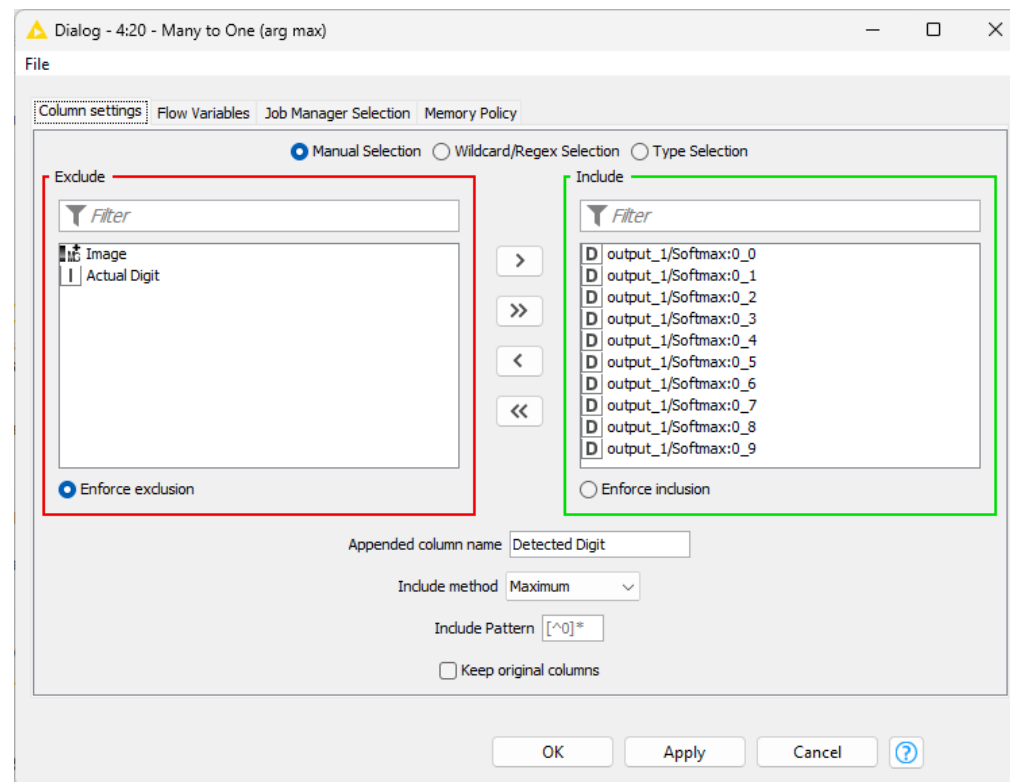
Part 3: Train and apply model



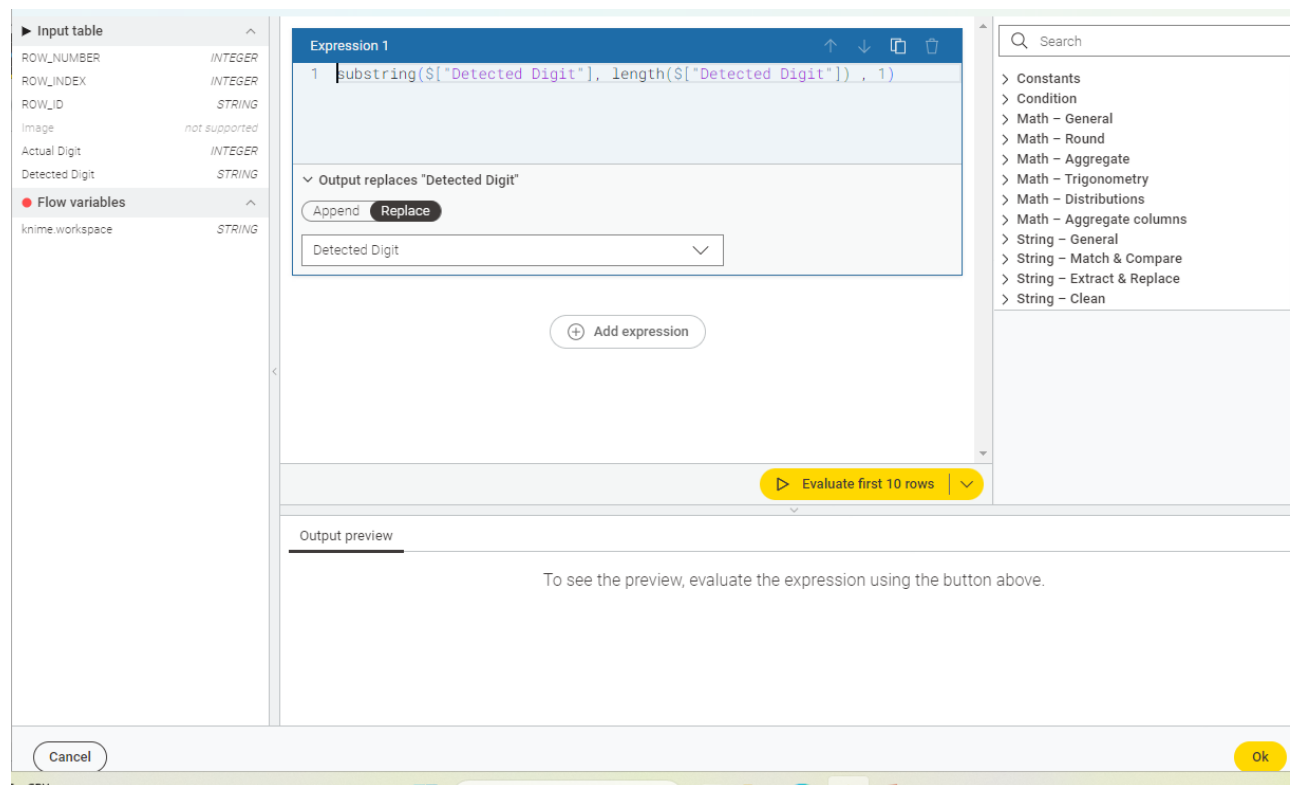
Part 4: Extract and evaluate trained network



Important Setting: Many to One



Important Setting: Expression



Convolution Neural Networks (CNN)


Confusion Matrix - 7:101 - Scorer

—
□
✕

File

Hilite

Actual Digi...	7	2	1	0	4	9	5	6	3	8
7	132	7	2	0	0	7	0	0	1	1
2	2	156	0	0	0	1	0	2	1	10
1	0	0	178	0	0	0	0	0	0	1
0	0	0	0	125	0	0	0	1	0	0
4	0	0	0	0	154	8	0	3	0	2
9	0	0	1	1	1	138	0	0	0	3
5	1	0	0	1	0	2	127	1	2	3
6	1	0	1	3	1	0	2	121	0	2
3	4	0	0	1	0	1	4	1	139	4
8	1	0	1	1	2	0	1	0	2	132

Correct classified: 1,402

Accuracy: 93.467%

Cohen's kappa (κ): 0.927%

Wrong classified: 98

Error: 6.533%